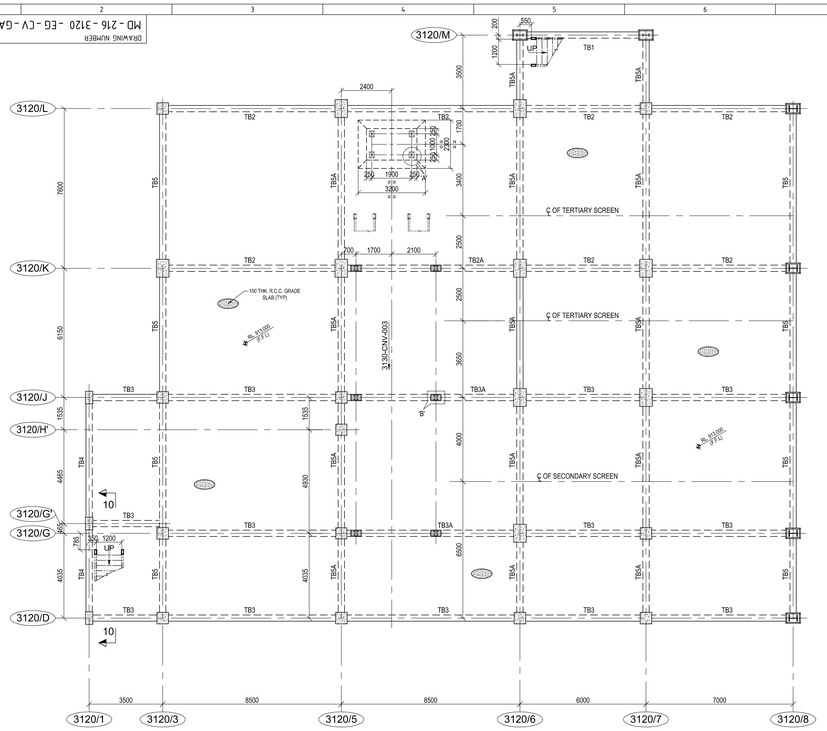
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	03-09-2020	14-10-2020	31-10-2023				Date	03-09-2020	14-10-2020	31-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

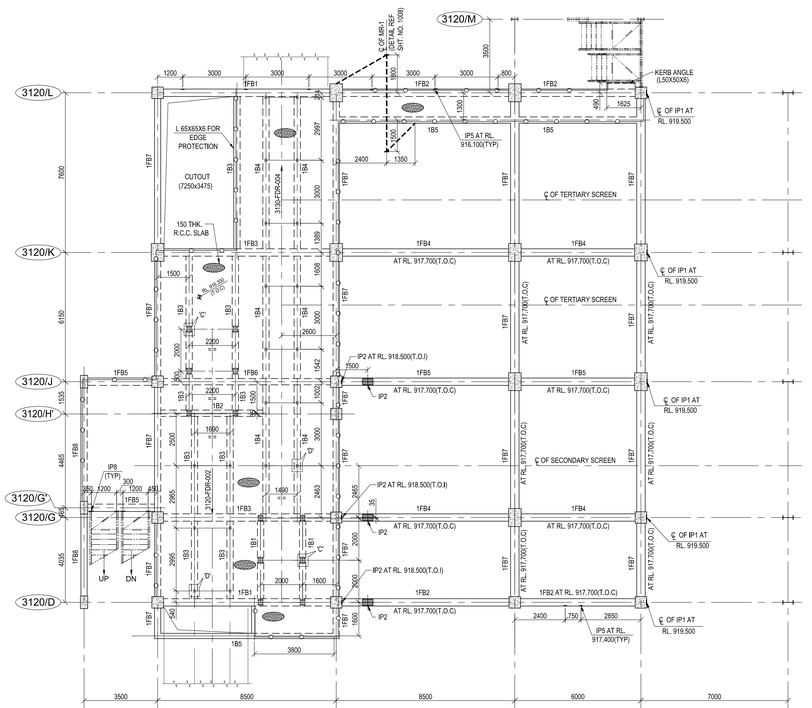
ATTACHMENT 3 **RELATED CIVIL GA & RC DRAWINGS OF NEARBY STRUCTURE**

DRAWING NUMBER: MD-216-3120-EG-CV-GAD-1005 REV D01



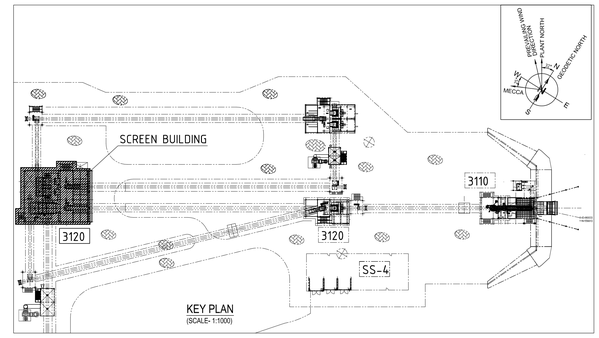
PLAN OF GRADE SLAB AT RL.913.000 (F.F.L) AND TIE BEAM AT RL. 912.850 (T.O.C)

(SCALE 1:100)



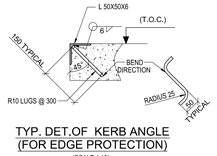
PLAN AT RL. 916.300(T.O.C)

(SCALE 1:100)



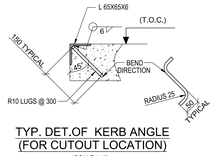
KEY PLAN

(SCALE- 1:1000)



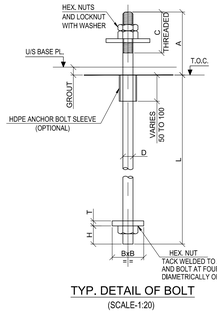
TYP. DET.OF KERB ANGLE (FOR EDGE PROTECTION)

(SCALE 1:10)



TYP. DET.OF KERB ANGLE (FOR CUTOUT LOCATION)

(SCALE 1:10)



TYP. DETAIL OF BOLT

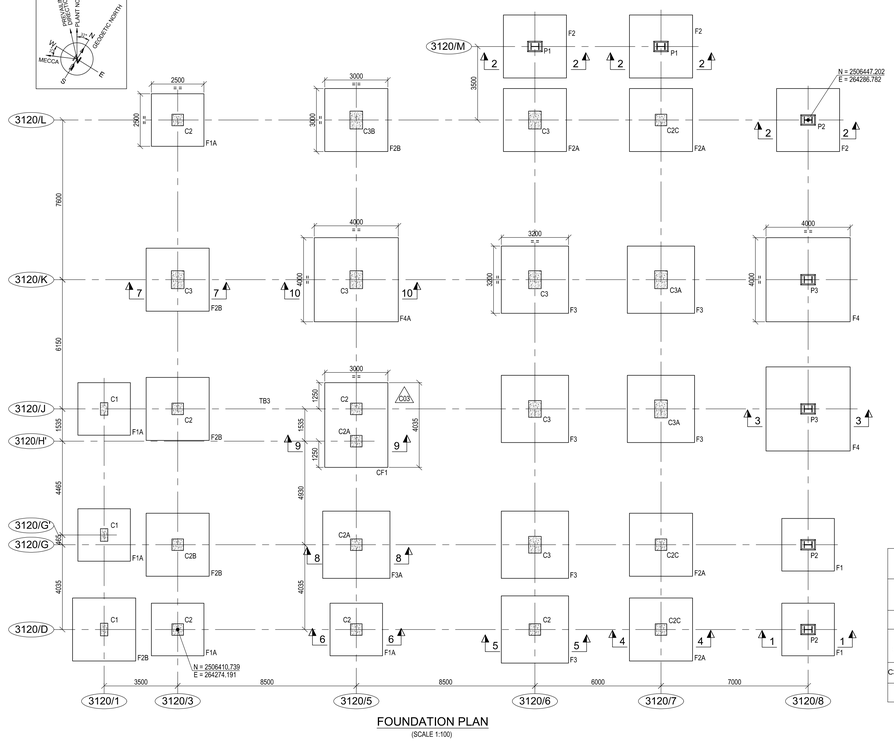
(SCALE-1:20)

SCHEDULE OF BOLTS (GRADE - A36)

D BOLT DIA. MM	BxB BOTTOM PLATE (SQUARE) SQ MM	DIMENSIONS (MM)				
		L	A	C	T	H
16	50x50	350	150	100	10	30
20	60x60	450	160	110	10	30
25	75x75	600	175	125	12	30
32	100x100	800	190	140	16	35
40	120x120	1000	225	175	20	45

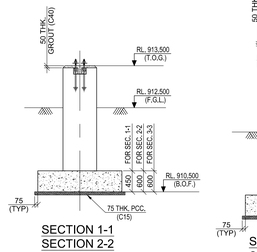
SCREEN FIXING BOLT OF GRADE - 8.8

D BOLT DIA. MM	BxB BOTTOM PLATE (SQUARE) SQ MM	DIMENSIONS (MM)				
		L	A	C	T	H
16	50x50	-	125	100	12	30



FOUNDATION PLAN

(SCALE 1:100)



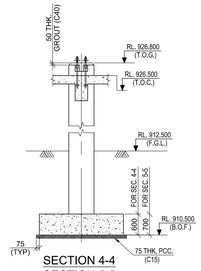
SECTION 1-1

SECTION 2-2

SECTION 3-3

(FOR FOUNDATION MKD, F1, F2 & F4)

(SCALE - 1:50)

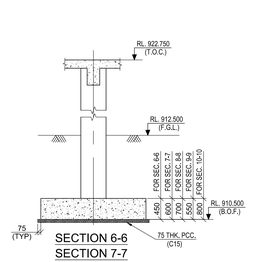


SECTION 4-4

SECTION 5-5

(FOR FOUNDATION MKD, F2A & F3)

(SCALE - 1:50)



SECTION 6-6

SECTION 7-7

SECTION 8-8

SECTION 9-9

SECTION 10-10

(FOR FOUNDATION MKD, F1A, F2B, F3A, F4A & CF1)

(SCALE - 1:50)

SCHEDULE OF FOUNDATION

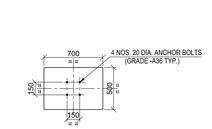
FOUNDATION MKD.	SIZE			BOTTOM OF FOUNDATION
	WIDTH (mm.)	LENGTH (mm.)	THICKNESS (mm.)	
F1 & F1A	2500	2500	450	RL. 910.500
F2, F2A & F2B	3000	3000	600	RL. 910.500
F3 & F3A	3200	3200	700	RL. 910.500
F4	4000	4000	600	RL. 910.500
CF1	3000	4035	650	RL. 910.500
F4A	4000	4000	800	RL. 910.500

SCHEDULE OF COLUMN

COLUMN MKD.	SIZE	
	WIDTH (mm.)	LENGTH (mm.)
C1	350	600
C2,C2A, C2B & C2C	550	550
C3, C3A & C3B	600	850
P1, P2, P3 & P4	500	700

SCHEDULE OF BEAM

BEAM MKD.	SIZE		LEVEL
	WIDTH (mm.)	DEPTH (mm.)	
TB1, TB2 , TB3, TB2A & TB3A	300	550	RL. 912.850 (T.O.B.)
TB4, TB5 & TB5A	300	500	
1FB1	350	850	RL. 916.300 (T.O.C.) RL. 917.700 (T.O.C.)
1FB2,1FB4 & 1FB5	350	600	
1FB3 & 1FB6	350	1000	
1FB7 &1FB8	300	550	
1B1 & 1B2	300	500	
1B3,1B4 & 1B5	300	500	

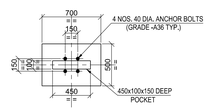


DETAIL OF COLUMN BASE FOR PEDASTAL MKD. P1

(SCALE-1:25)

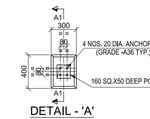
DETAIL OF COLUMN BASE FOR PEDASTAL MKD. P2

(SCALE-1:25)



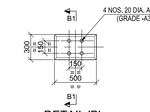
DETAIL OF COLUMN BASE FOR PEDASTAL MKD. P3

(SCALE-1:25)



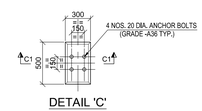
DETAIL - 'A'

(SCALE-1:25)



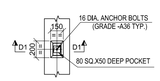
DETAIL 'B'

(SCALE-1:25)



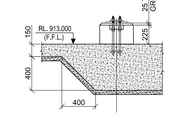
DETAIL 'C'

(SCALE-1:25)



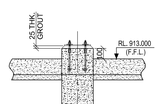
DETAIL 'D'

(SCALE-1:25)



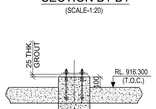
SECTION A1-A1

(SCALE-1:25)



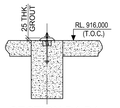
SECTION B1-B1

(SCALE-1:20)



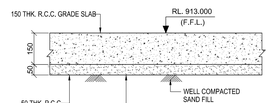
SECTION C1-C1

(SCALE-1:20)



SECTION D1-D1

(SCALE-1:20)



TYP. DET. OF 150 THK. GRADE SLAB

(SCALE 1:10)

LEGENDS :-

RL - REDUCED LEVEL
℄ - CENTRE LINE
T.O.C. - TOP OF CONCRETE.
F.G.L. - FINISHED GROUND LEVEL.
B.O.F. - BOTTOM OF FOUNDATION
B.O.B. - BOTTOM OF BEAM
TYP. - TYPICAL
R.C.C - REINFORCED CEMENT CONCRETE
P.C.C - PLAIN CEMENT CONCRETE
MKD. - MARKED
THK. - THICK
MR. - MONORAIL
B.O.M.R. - BOTTOM OF MONORAIL
SEC. - SECTION
IP - INSERT PLATE
MM - MILLIMETER
SQ. - SQUARE
T.O.B. - TOP OF BEAM
T.O.I. - TOP OF INSERT

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STAIETD.
2. GRADE OF CONCRETE SHALL BE C35 FOR FOUNDATION.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615 - GRADE 60 (fy = 420 MPa).
4. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDERED 500KN/SQ. M.
5. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS
   a) FOUNDATION = 75mm (BOTTOM & SIDE) 50 mm (TOP)
   b) TIE BEAM = 50 mm c) COLUMN = 50mm d) BEAM = 50mm e) SLAB = 40mm
6. FOR MINIMUM LAP LENGTH OF REINFORCEMENT REFER STANDARD DRAWING NO. MD-216-1000-EG-CV-DED-1001
7. THIS DRG. SHALL BE READ IN CONJUNCTION WITH DRG. NO. MD-216-3120-EG-CV-GAD-1006 & 1008, MD-216-3120-EG-CV-DED-1005 & 1006
8. FOR DETAIL OF HANDRAIL FIXING REFER STANDARD DRAWING.

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3120-EG-ME-LPX-1004-RA01	SCREEN HOUSE-PLAN VIEW
MD-216-3120-EG-ME-LPX-1005-RA01	SCREEN HOUSE - PLAN VIEW
MD-216-3120-EG-ME-LPX-1006-RA01	SCREEN HOUSE - PLAN VIEW
MD-216-3120-EG-ME-LPX-1007-RA01	SCREEN HOUSE - PLAN VIEW
MD-216-3120-EG-ME-LPX-1008-RA01	SCREEN HOUSE - PLAN VIEW
MD-216-3120-EG-ME-ELV-1008-A01	SCREEN HOUSE - ELEVATION
MD-216-3120-EG-ME-LPX-1009-RA01	SCREEN HOUSE - PLAN VIEW
MD-216-3120-EG-ME-ELV-1009-RA01	SCREEN HOUSE - PLAN VIEW
MD-216-3120-EG-ME-ELV-1010-RA01	SCREEN HOUSE - ELEVATION

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
D01	31/10/2023	AS BUILT	JKM	SBY	RPK	AG	
C03	12/08/2020	ISSUED FOR CONSTRUCTION & REVISED AS MKD.	JKM	SBY	RPK	AG	
C02	22/06/2020	ISSUED FOR CONSTRUCTION & REVISED AS MKD.	MGH	SBY	RPK	AG	
C01	09/04/2020	ISSUED FOR CONSTRUCTION, REVISED AS PER EMPLOYER COMMENTS DATED 04/04/2020 & REVISED AS MKD.	MGH	SBY	RPK	AG	
A01	07/03/2020	ISSUED OF REFERENCE	MGH	SBY	RPK	AG	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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LARSEN & TOUBRO DRAWING NUMBER: 019045_100_C_G_24068_001 — REVISION: 4

OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31258 — DOCUMENT ID: OU100670457 — REVISION: 4

DRAWN	MGH	29/02/2020
CHECKED	SBY	05/03/2020
ENGINEER	RPK	04/03/2020
APPROVED	AG	07/03/2020
EMPLOYER		
SCALE	1:100,1:75,1:50,1:25,1:20,1:10	A0

MANSOURAH - MASSARAH GOLD PROJECT

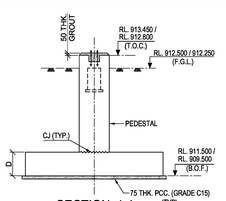
GENERAL ARRANGEMENT DRAWING

SCREEN BUILDING

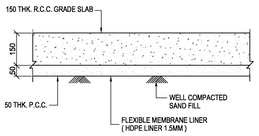
FOUNDATION LAYOUT, FLOOR PLAN & DETAILS

DRAWING NUMBER: MD - 216 - 3120 - EG - CV - GAD - 1005 — REV: D01

DRAWING NUMBER: MD-216-3130-EG-CV-GAD-1003 REV D01



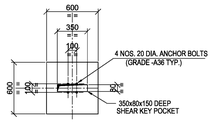
SECTION 1-1
(SCALE-1:50)



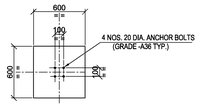
TYP. DETAIL OF 150 THK. GRADE SLAB
SCALE 1:10

SCHEDULE OF BEAM

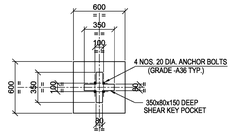
BEAM MKD.	SIZE WIDTH (mm.)	SIZE DEPTH (mm.)	LEVEL
TB1	300	500	RL. 912.500 (T.O.B.)
TB1A	300	600	RL. 912.500 (T.O.B.)
TB2	300	500	RL. 912.500 (T.O.B.)
TB3	300	500	RL. 912.500 (T.O.B.)
TB4	300	500	RL. 912.850 (T.O.B.)
TB5	300	500	RL. 912.850 (T.O.B.)
TB6	300	600	RL. 912.850 (T.O.B.)
TB7	300	600	RL. 912.500 (T.O.B.)
TB8	300	700	RL. 912.500 (T.O.B.)



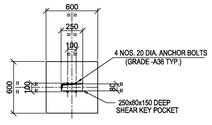
DETAIL OF PEDESTAL MKD. P1
(SCALE-1:25)



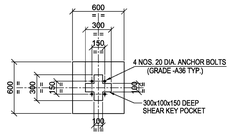
DETAIL OF PEDESTAL MKD. P2 & P6
(SCALE-1:25)



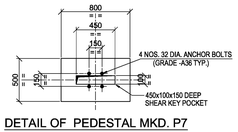
DETAIL OF PEDESTAL MKD. P3
(SCALE-1:25)



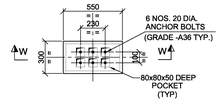
DETAIL OF PEDESTAL MKD. P4
(SCALE-1:25)



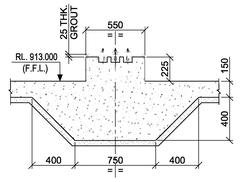
DETAIL OF PEDESTAL MKD. P5
(SCALE-1:25)



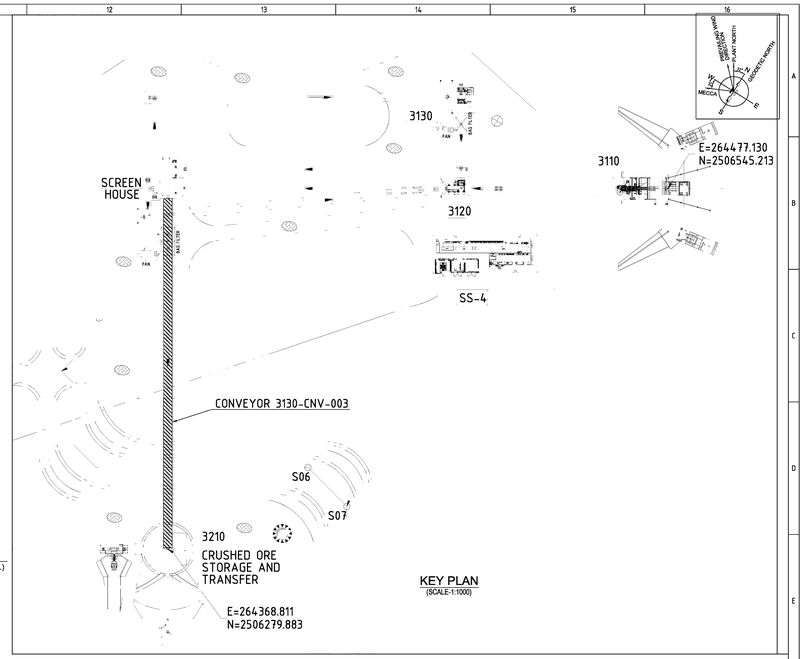
DETAIL OF PEDESTAL MKD. P7
(SCALE-1:25)



DETAIL OF PEDESTAL MKD. P9
(SCALE-1:20)



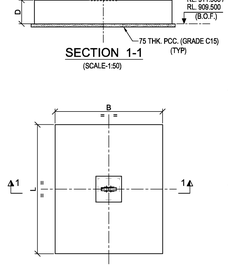
SECTION W-W
(SCALE-1:20)



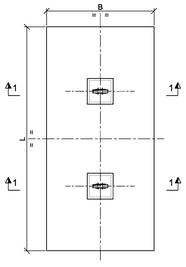
KEY PLAN
(SCALE-1:1000)

SCHEDULE OF FOUNDATION

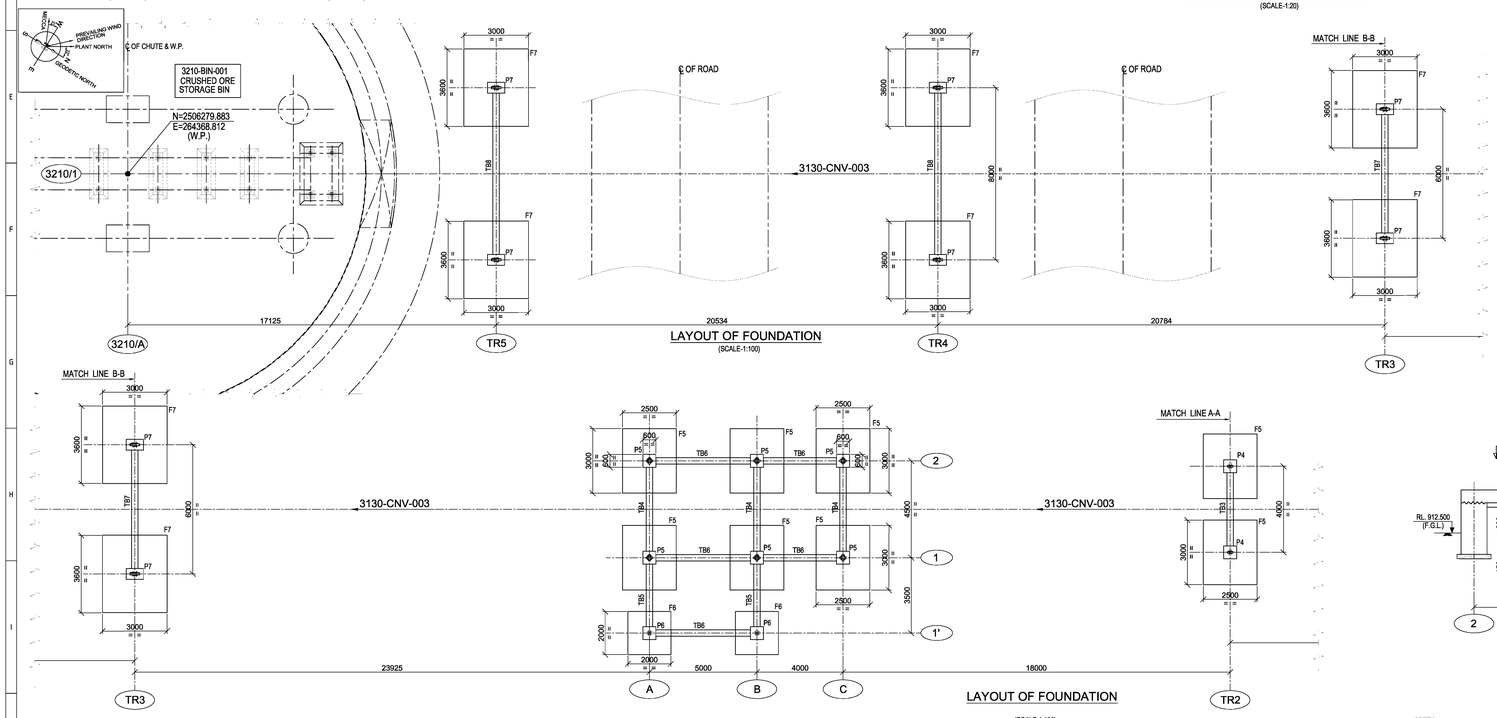
FDN. MKD.	TYPE MKD.	PEDESTAL	FOUNDATION DETAILS L	B	D	BOTTOM OF FDN.	TOP OF PEDESTAL	NOS. FDN.
F1	I	P1	2000	1500	300	RL. 910.500	RL. 912.800	2
F2	I	P1 & P2	2000	1500	300	RL. 911.500	RL. 912.800	6
F3	I	P3	2000	2000	300	RL. 911.500	RL. 912.800	4
F4	II	P4	4000	3000	450	RL. 909.500	RL. 912.800	1
F5	I	P4 & P5	3000	2500	450	RL. 909.500	RL. 912.800 FOR P4 RL. 913.450 FOR P5	8
F6	I	P6	2000	2000	350	RL. 909.500	RL. 913.450	2
F7	I	P7	3600	3000	450	RL. 909.500	RL. 912.800	6

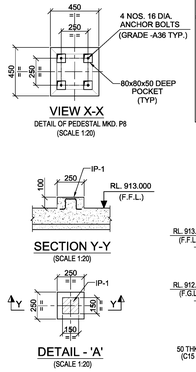


TYPICAL FOUNDATION PLAN
(TYPE-I)
(SCALE-1:50)



TYPICAL FOUNDATION PLAN
(TYPE-II)
(SCALE-1:50)

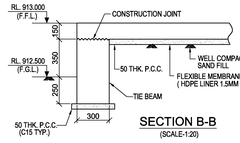




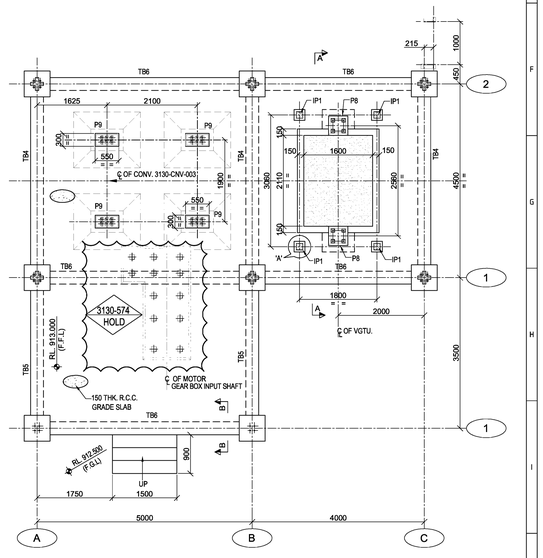
VIEW X-X
DETAIL OF PEDESTAL MKD. P8
(SCALE 1:20)

SECTION Y-Y
(SCALE 1:20)

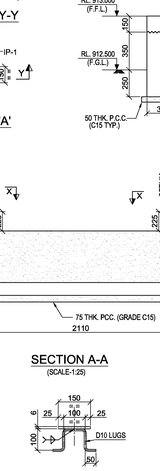
DETAIL - 'A'
(SCALE 1:20)



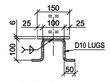
SECTION B-B
(SCALE-1:20)



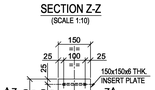
PLAN OF GRADE SLAB AT RL. 913.000 AND TIE BEAM AT RL. 912.850 (T.O.B) (FOR VGTU TOWER)
(SCALE-1:50)



SECTION A-A
(SCALE-1:25)



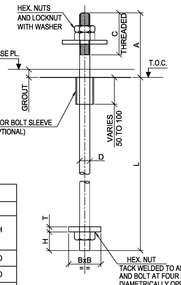
SECTION Z-Z
(SCALE 1:10)



DETAIL OF INSERT PLATE - IP1
(SCALE-1:10)

SCHEDULE OF BOLTS

D BOLT DIA. MM	BxB BOTTOM PLATE (SQUARE) SQ MM	DIMENSIONS (MM) L	A	C	T	H
16	50x50	350	120	80	10	30
20	60x60	450	160	110	10	30
32	100x100	800	200	150	16	35



TYP. DETAIL OF BOLT

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. GRADE OF CONCRETE SHALL BE C35 FOR FOUNDATION.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615 - GRADE 60 (fy = 420 MPa).
4. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDERED 500KN/ SQ. M.
5. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS
   a) FOUNDATION = 75 mm (BOTTOM & SIDE) 50 mm. (TOP)
   b) COLUMN & PEDESTAL = 50mm
   c) BEAM = 50mm
   d) SLAB = 40mm
6. FOR MINIMUM LAP LENGTH OF REINFORCEMENT AND OTHER GENERAL NOTES FOR CIVIL WORKS REFER STANDARD DRG : MD-216-1000-EG-CV-DED-1001
7. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD - 216 - 3130 -EG -ST - GAD - 1011 TO MD - 216 - 3130 -EG -ST - GAD - 1016
8. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO. MD - 216 - 1000 -EG -ST - CRT - 1001
9. FOR OTHER STANDARD REINFORCEMENT DETAILS REFER DRG : MD-216-1000-EG-CV-DED-1005

LEGENDS :-

RL. - REDUCED LEVEL
℄ - CENTRE LINE
CJ - CONSTRUCTION JOINT
FDN. - FOUNDATION.
T.O.C. - TOP OF CONCRETE.
F.G.L. - FINISHED GROUND LEVEL.
F.F.L. - FINISHED FLOOR LEVEL.
B.O.F. - BOTTOM OF FOUNDATION
TYP. - TYPICAL
B.O.B. - BOTTOM OF BEAM.
B.O.M. - BOTTOM OF MONORAIL.
R.C.C. - REINFORCED CEMENT CONCRETE
P.C.C. - PLAIN CEMENT CONCRETE
IP. - INSERT PLATE
MKD. - MARKED.
THK. - THICK

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-ELV-1011	TERTIARY CRUSHING CONV.3130-CNV-003 - ELEVATION	D01	31/10/2023	AS BUILT	AMJ	SBY	RPK/APL	PRI	
MD-216-3130-EG-ME-ELV-1010	TERTIARY CRUSHING CONV.3130-CNV-003 - ELEVATION	C01	29/05/2020	ISSUED FOR CONSTRUCTION, REVISED AS PER EMPLOYER	AMJ	SBY	RPK/APL	PRI	
MD-216-3130-EG-ME-ELV-1009	TERTIARY CRUSHING CONV.3130-CNV-003 - ELEVATION			COMMENTS DATED 26/05/2020 & REVISED AS MKD. 1.01					
MD-216-3130-EG-ME-LPX-1005	TERTIARY CRUSHING CONV.3130-CNV-003 - PLAN	A01	30/03/2020	ISSUED FOR REFERENCE	AMJ	SBY	RPK/APL	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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LARSEN & TOUBRO DRAWING NUMBER	O19045_100_C_G_24056_006
REVISION	2
OUTOTEC DRAWING NUMBER	900707 _ GAU01 _ ZC01 _ MTL01 _ 31359
DOCUMENT ID	OU100670366
REVISION	2
DRAWN	AMJ 28/03/2020
CHECKED	SBY 29/03/2020
ENGINEER	RPK/APL 27/03/2020
APPROVED	PRI 30/03/2020
EMPLOYER	
SCALE	1:100,50,20,10 A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GA OF CONVEYOR 3130-CNV-003
LAYOUT & DETAIL

DRAWING NUMBER: MD - 216 - 3130 - EG - CV - GAD - 1003
REV: D01

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	03-09-2020	14-10-2020	31-10-2023				Date	03-09-2020	14-10-2020	31-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 4 **STAAD INPUT OF TOTAL FOUNDATION ARRANGEMENT**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 28-Aug-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 -13.39 -3.2 -10.59; 2 -12.865 -3.2 -10.59; 3 -12.34 -3.2 -10.59;
4 -11.89 -3.2 -10.59; 5 -11.44 -3.2 -10.59; 6 -10.915 -3.2 -10.59;
7 -10.39 -3.2 -10.59; 8 -9.89 -3.2 -10.49; 9 -9.39 -3.2 -10.49;
10 -8.89 -3.2 -10.49; 11 -8.49 -3.2 -10.49; 12 -7.89 -3.2 -10.49;
13 -7.39 -3.2 -10.49; 14 -6.89 -3.2 -10.49; 15 -6.39 -3.2 -10.49;
16 -9.89 -3.2 -10.09; 17 -9.39 -3.2 -10.09; 18 -8.89 -3.2 -10.09;
19 -8.49 -3.2 -10.09; 20 -7.89 -3.2 -10.09; 21 -7.39 -3.2 -10.09;
22 -6.89 -3.2 -10.09; 23 -6.39 -3.2 -10.09; 24 -3.09 -3.2 -10.09;
25 -2.59 -3.2 -10.09; 26 -2.09 -3.2 -10.09; 27 -1.59 -3.2 -10.09;
28 -1.09 -3.2 -10.09; 29 -0.59 -3.2 -10.09; 30 -0.09 -3.2 -10.09;
31 0.41 -3.2 -10.09; 32 0.91 -3.2 -10.09; 33 -13.39 -3.2 -10.065;
34 -12.865 -3.2 -10.065; 35 -12.34 -3.2 -10.065; 36 -11.89 -3.2 -10.065;
37 -11.44 -3.2 -10.065; 38 -10.915 -3.2 -10.065; 39 -10.39 -3.2 -10.065;
40 -9.89 -3.2 -9.59; 41 -9.39 -3.2 -9.59; 42 -8.89 -3.2 -9.59;
43 -8.49 -3.2 -9.59; 44 -7.89 -3.2 -9.59; 45 -7.39 -3.2 -9.59;
46 -6.89 -3.2 -9.59; 47 -6.39 -3.2 -9.59; 48 -3.09 -3.2 -9.59;
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61 -11.44 -3.2 -9.54; 62 -10.915 -3.2 -9.54; 63 -10.39 -3.2 -9.54;
64 -13.39 -3.2 -9.09; 65 -12.865 -3.2 -9.09; 66 -12.34 -3.2 -9.09;
67 -11.89 -3.2 -9.09; 68 -11.44 -3.2 -9.09; 69 -10.915 -3.2 -9.09;
70 -10.39 -3.2 -9.09; 71 -9.89 -3.2 -9.09; 72 -9.39 -3.2 -9.09;
73 -8.89 -3.2 -9.09; 74 -8.49 -3.2 -9.09; 75 -7.89 -3.2 -9.09;
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85 -0.09 -3.2 -9.09; 86 0.41 -3.2 -9.09; 87 0.91 -3.2 -9.09;
88 -13.39 -3.2 -8.64; 89 -12.865 -3.2 -8.64; 90 -12.34 -3.2 -8.64;
91 -11.89 -3.2 -8.64; 92 -11.44 -3.2 -8.64; 93 -10.915 -3.2 -8.64;
94 -10.39 -3.2 -8.64; 95 -9.89 -3.2 -8.59; 96 -9.39 -3.2 -8.59;
97 -8.89 -3.2 -8.59; 98 -8.49 -3.2 -8.59; 99 -7.89 -3.2 -8.59;
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115 -11.89 -3.2 -8.115; 116 -11.44 -3.2 -8.115; 117 -10.915 -3.2 -8.115;
118 -10.39 -3.2 -8.115; 119 -9.89 -3.2 -8.09; 120 -9.39 -3.2 -8.09;
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133 -0.09 -3.2 -8.09; 134 0.41 -3.2 -8.09; 135 0.91 -3.2 -8.09;
136 -13.39 -3.2 -7.59; 137 -12.865 -3.2 -7.59; 138 -12.34 -3.2 -7.59;
139 -11.89 -3.2 -7.59; 140 -11.44 -3.2 -7.59; 141 -10.915 -3.2 -7.59;
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142 -10.39 -3.2 -7.59; 143 -9.89 -3.2 -7.59; 144 -9.39 -3.2 -7.59;
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148 -7.39 -3.2 -7.59; 149 -6.89 -3.2 -7.59; 150 -6.39 -3.2 -7.59;
151 -12.89 -3.2 -5.09; 152 -12.34 -3.2 -5.09; 153 -11.89 -3.2 -5.09;
154 -11.44 -3.2 -5.09; 155 -10.89 -3.2 -5.09; 156 -8.89 -3.2 -4.84;
157 -8.49 -3.2 -4.84; 158 -7.89 -3.2 -4.84; 159 -7.39 -3.2 -4.84;
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172 -11.89 -3.2 -4.09; 173 -11.44 -3.2 -4.09; 174 -10.89 -3.2 -4.09;
175 -8.89 -3.2 -4.09; 176 -8.49 -3.2 -4.09; 177 -7.89 -3.2 -4.09;
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187 0.41 -3.2 -4.09; 188 0.91 -3.2 -4.09; 189 1.41 -3.2 -4.09;
190 1.91 -3.2 -4.09; 191 2.41 -3.2 -4.09; 192 2.91 -3.2 -4.09;
193 3.41 -3.2 -4.09; 194 14.86 -3.2 -3.99; 195 15.56 -3.2 -3.99;
196 16.06 -3.2 -3.99; 197 16.56 -3.2 -3.99; 198 17.26 -3.2 -3.99;
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226 16.56 -3.2 -3.29; 227 17.26 -3.2 -3.29; 228 -1.09 -3.2 -3.24;
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271 14.86 -3.2 -2.59; 272 15.56 -3.2 -2.59; 273 16.06 -3.2 -2.59;
274 16.56 -3.2 -2.59; 275 17.26 -3.2 -2.59; 276 -14.47 -3.2 -2.565;
277 -13.82 -3.2 -2.565; 278 -13.22 -3.2 -2.565; 279 -12.72 -3.2 -2.565;
280 -12.22 -3.2 -2.565; 281 -11.62 -3.2 -2.565; 282 -10.97 -3.2 -2.565;
283 -20.25 -3.2 -2.315; 284 -19.75 -3.2 -2.315; 285 -19.25 -3.2 -2.315;
286 -18.75 -3.2 -2.315; 287 -18.25 -3.2 -2.315; 288 -17.75 -3.2 -2.315;
289 -17.25 -3.2 -2.315; 290 -8.89 -3.2 -2.315; 291 -8.49 -3.2 -2.315;
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MEMBER INCIDENCES

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926 44 45 21 20; 927 75 76 45 44; 928 99 100 76 75; 929 123 124 100 99;
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970 133 134 110 109; 971 55 56 32 31; 972 86 87 56 55; 973 110 111 87 86;
974 134 135 111 110;
START GROUP DEFINITION
ELEMENT
_MX 203 208 213 218 223 241 242 245 TO 248 251 TO 254 257 TO 260 263 TO 266 -
269 TO 272 275 TO 277 280 281 284 285 288 289 292 321 TO 330 371 TO 380 401 -
402 TO 421 444 TO 449 452 TO 454 521 522 525 TO 528 531 TO 534 537 TO 540 -
543 TO 546 549 TO 552 555 TO 558 561 TO 564 567 TO 570 573 TO 576 -
579 TO 582 585 TO 588 591 592 601 602 605 TO 608 611 TO 614 617 TO 620 623 -
624 TO 626 629 TO 632 635 TO 638 641 TO 644 647 TO 650 653 TO 656 659 TO 662 -
665 TO 668 671 672 701 TO 714 729 TO 742 805 TO 811 816 TO 822 856 TO 866 -
901 TO 912 937 TO 946 955 TO 962 971 TO 974
_MY 201 TO 210 216 TO 220 241 TO 252 265 TO 280 289 TO 292 321 325 326 329 -
330 TO 331 335 336 339 TO 341 345 346 349 TO 351 355 356 360 361 365 366 370 -
371 TO 372 375 376 380 TO 382 385 386 390 TO 392 395 396 399 TO 402 405 406 -
409 TO 411 413 TO 415 419 TO 422 424 TO 426 430 TO 433 435 TO 437 -
441 TO 444 446 TO 448 452 TO 454 521 TO 532 545 TO 568 581 TO 592 -
601 TO 612 625 TO 648 661 TO 672 701 702 705 TO 709 712 TO 716 719 TO 723 -
726 TO 730 733 TO 737 740 TO 742 805 TO 807 810 811 816 TO 818 821 822 827 -
828 TO 829 832 833 838 TO 840 843 TO 846 849 TO 851 854 TO 857 860 TO 862 -
865 TO 868 871 872 901 902 905 TO 908 911 TO 914 917 TO 920 923 TO 926 929 -
930 TO 932 935 TO 938 941 TO 943 946 947 950 951 954 955 958 959 962 963 966 -
967 970 971 974
_SHEAR 201 TO 205 241 TO 247 252 253 258 259 264 265 270 TO 277 280 289 292 -
325 TO 330 335 TO 340 390 400 410 411 420 TO 422 431 TO 433 442 TO 444 453 -
454 522 TO 525 527 532 533 538 539 544 545 550 552 TO 555 558 TO 561 563 -
568 569 574 575 580 581 586 588 TO 591 602 TO 605 607 612 613 618 619 624 -
625 630 632 TO 635 638 TO 641 643 648 649 654 655 660 661 666 668 TO 671 -
702 TO 706 708 714 715 721 722 728 729 735 737 TO 741 804 TO 811 817 TO 822 -
834 845 856 867 871 901 TO 907 912 913 918 919 924 925 930 931 936 937 942
_MX_700 601 602 605 TO 608 611 TO 614 617 TO 620 623 TO 626 629 TO 632 635 -
636 TO 638 641 TO 644 647 TO 650 653 TO 656 659 TO 662 665 TO 668 671 672 -
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701 TO 714 729 TO 742 805 TO 811 816 TO 822 856 TO 866
_MX_400 203 208 213 218 223 277 280 281 284 285 288 289 292
_MY_700 601 TO 612 625 TO 648 661 TO 672 701 702 705 TO 709 712 TO 716 719 -
720 TO 723 726 TO 730 733 TO 737 740 TO 742 805 TO 807 810 811 816 TO 818 -
821 822 827 TO 829 832 833 838 TO 840 843 TO 846 849 TO 851 854 TO 857 860 -
861 TO 862 865 TO 868 871 872
_MY_400 201 TO 210 216 TO 220 277 TO 280 289 TO 292
_SHEAR_700 602 TO 605 607 612 613 618 619 624 625 630 632 TO 635 638 TO 641 -
643 648 649 654 655 660 661 666 668 TO 671 702 TO 706 708 714 715 721 722 -
728 729 735 737 TO 741 804 TO 811 817 TO 822 834 845 856 867 871
_SHEAR_400 201 TO 205 277 280 289 292
_MX_600 241 242 245 TO 248 251 TO 254 257 TO 260 263 TO 266 269 TO 272 275 -
276 321 TO 330 371 TO 380 401 TO 421 444 TO 449 452 TO 454 521 522 -
525 TO 528 531 TO 534 537 TO 540 543 TO 546 549 TO 552 555 TO 558 -
561 TO 564 567 TO 570 573 TO 576 579 TO 582 585 TO 588 591 592 901 TO 912 -
937 TO 946 955 TO 962 971 TO 974
_MY_600 241 TO 252 265 TO 276 321 325 326 329 TO 331 335 336 339 TO 341 345 -
346 349 TO 351 355 356 360 361 365 366 370 TO 372 375 376 380 TO 382 385 -
386 390 TO 392 395 396 399 TO 402 405 406 409 TO 411 413 TO 415 419 TO 422 -
424 TO 426 430 TO 433 435 TO 437 441 TO 444 446 TO 448 452 TO 454 -
521 TO 532 545 TO 568 581 TO 592 901 902 905 TO 908 911 TO 914 917 TO 920 -
923 TO 926 929 TO 932 935 TO 938 941 TO 943 946 947 950 951 954 955 958 959 -
962 963 966 967 970 971 974
_SHEAR_600 241 TO 247 252 253 258 259 264 265 270 TO 276 325 TO 330 -
335 TO 340 390 400 410 411 420 TO 422 431 TO 433 442 TO 444 453 454 -
522 TO 525 527 532 533 538 539 544 545 550 552 TO 555 558 TO 561 563 568 -
569 574 575 580 581 586 588 TO 591 901 TO 907 912 913 918 919 924 925 930 -
931 936 937 942
END GROUP DEFINITION
ELEMENT PROPERTY
601 TO 672 701 TO 742 801 TO 874 THICKNESS 0.7
241 TO 276 321 TO 454 521 TO 592 901 TO 974 THICKNESS 0.6
201 TO 225 THICKNESS 0.4
277 TO 292 THICKNESS 0.4
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 35000
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 12 PRIS YD 0.9 ZD 1.2
21 TO 32 PRIS YD 0.6 ZD 0.6
41 TO 48 PRIS YD 0.6 ZD 0.6
51 TO 62 PRIS YD 0.7 ZD 0.5
49 50 PRIS YD 0.55 ZD 0.55
63 TO 70 PRIS YD 0.45 ZD 0.45
*
82 TO 85 89 90 93 TO 95 PRIS YD 0.6 ZD 0.3
102 TO 106 PRIS YD 0.4 ZD 0.3
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98 TO 101 PRIS YD 0.3 ZD 0.3
MEMBER PROPERTY
81 86 TO 88 91 92 96 97 PRIS YD 0.3 ZD 0.5
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
1 TO 727 ELASTIC MAT DIRECT Y SUBGRADE 30000
MEMBER OFFSET
63 TO 70 100 101 105 106 START 0 0 -0.225
63 TO 70 100 101 105 106 END 0 0 -0.225
LOAD 1 LOADTYPE Dead  TITLE DEAD LOAD
SELFWEIGHT Y -1
ELEMENT LOAD
201 TO 225 PR GY -35.1
241 TO 276 321 TO 454 521 TO 592 901 TO 974 PR GY -63.7
277 TO 292 601 TO 672 701 TO 742 801 TO 874 PR GY -60.9
JOINT LOAD
**Duct Support Structure
757 FX -0.1 FY -57 FZ 0
758 FX 0.1 FY -57 FZ 0
760 FX 0.1 FY -3.4 FZ 0
759 FX -0.1 FY -3.4 FZ 0
756 FX 0.1 FY -201 FZ -0.2
763 FX 0.1 FY -202 FZ 0.2
754 FX -0.2 FY -139 FZ -0.2
755 FX 0.1 FY -118 FZ -0.1
762 FX 0.1 FY -124 FZ 0.2
761 FX -0.1 FY -140 FZ 0.1
**** Conv 3130-003
764 FX 1.5 FY -18.2 FZ -3.2
765 FX -1.5 FY -25.9 FZ -5.9
**** TP-2
779 FX 4.4 FY -72 FZ 12.3
781 FX 0 FY -31.8 FZ -8.5
780 FX 1.4 FY -1.6 FZ -2.9
778 FX -0.8 FY -59.3 FZ 7
777 FX 0 FY 2.9 FZ -6.8
776 FX -1.2 FY -23.9 FZ -1.1
***Bag Filter
766 FX 0 FY -598 FZ -27
767 FX 0 FY -1131 FZ -51
768 FX 0 FY -607 FZ -28
769 FX 0 FY -582 FZ 27
770 FX 0 FY -1099 FZ 51
771 FX 0 FY -587 FZ 27
***StairCase
773 FX 4 FY -57.8 FZ -0.6
775 FX 0.9 FY -176.2 FZ 0.8
774 FX -0.8 FY -177.2 FZ 0.6
772 FX -4.1 FY -64.4 FZ -0.8
*********************************************************************************
LOAD 2 LOADTYPE Live  TITLE PIPE LOAD
JOINT LOAD
757 FX 0 FY 0 FZ 0
```

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758 FX 0 FY 0 FZ 0
760 FX 0 FY 0 FZ 0
759 FX 0 FY 0 FZ 0
756 FX 0.4 FY 0 FZ 0
763 FX 0.4 FY 0 FZ 0.1
754 FX -0.4 FY 0 FZ -0.1
755 FX -0.3 FY 0 FZ 0
762 FX 0 FY 0 FZ 0.1
761 FX -0.2 FY 0 FZ 0
****Conv 3130-003
764 FX -4.5 FY -33.9 FZ -5.1
765 FX 0.1 FY -43.2 FZ -18
779 FX 2.8 FY -64.1 FZ 5.6
781 FX 0 FY -27.5 FZ -6.3
780 FX 4.7 FY -26.6 FZ 0.9
778 FX -2.9 FY -45.7 FZ 2.1
777 FX 0 FY -26.7 FZ 1.5
776 FX -4.5 FY -57.7 FZ -3.8
766 FX 0.1 FY 0.1 FZ 0.1
767 FX 0.1 FY 0.1 FZ 0.1
768 FX 0.1 FY 0.1 FZ 0.1
769 FX 0.1 FY 0.1 FZ 0.1
770 FX 0 FY 0 FZ 0
771 FX 0 FY 0 FZ 0
773 FX 9.9 FY -114.7 FZ -0.1
775 FX 0.5 FY -102 FZ 0.2
774 FX -0.5 FY -115.8 FZ 0.1
772 FX -10 FY -132.5 FZ -0.2
*********************************************************************************
LOAD 3 LOADTYPE Wind  TITLE WIND LOAD (+X DIR)
JOINT LOAD
757 FX 6 FY 34.4 FZ -0.7
758 FX 5.7 FY -34.4 FZ 0.7
760 FX 4 FY -36.1 FZ 0.8
759 FX 4.2 FY 36.1 FZ -0.8
756 FX 4.3 FY -16.1 FZ -0.1
763 FX 4.3 FY -11.6 FZ -0.1
754 FX 21.9 FY 303.6 FZ 0.8
755 FX 22.8 FY -287.5 FZ -0.7
762 FX 25.1 FY -297.3 FZ -0.6
761 FX 24.1 FY 308.9 FZ 0.7
779 FX 16.1 FY -24.2 FZ 2.1
781 FX 2 FY 7.6 FZ -0.9
780 FX 4.9 FY -12.6 FZ -1.6
778 FX 16.1 FY 25.4 FZ -0.4
777 FX 2.1 FY -4.5 FZ 0.9
776 FX 6.8 FY 8.3 FZ 0
766 FX 4 FY 164 FZ 26
767 FX 171 FY 278 FZ 1
768 FX 3 FY -442 FZ -28
769 FX 7 FY 105 FZ 15
770 FX 426 FY 598 FZ 3
771 FX 2 FY -704 FZ -18
773 FX 47.9 FY -405.7 FZ -3.1
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775 FX 84 FY -328.2 FZ -2.5
774 FX -0.1 FY 328.2 FZ 2.4
772 FX 47.1 FY 405.7 FZ 3.3
****
LOAD 4 LOADTYPE Wind TITLE WIND LOAD (-X DIR)
JOINT LOAD
757 FX -6 FY -34.4 FZ 0.7
758 FX -5.7 FY 34.4 FZ -0.7
760 FX -4 FY 36.1 FZ -0.8
759 FX -4.2 FY -36.1 FZ 0.8
756 FX -4.3 FY 16.1 FZ 0.1
763 FX -4.3 FY 11.6 FZ 0.1
754 FX -21.9 FY -303.6 FZ -0.8
755 FX -22.8 FY 287.5 FZ 0.7
762 FX -25.1 FY 297.3 FZ 0.6
761 FX -24.1 FY -308.9 FZ -0.7
779 FX -16.1 FY 24.2 FZ -2.1
781 FX -2 FY -7.6 FZ 0.9
780 FX -4.9 FY 12.6 FZ 1.6
778 FX -16.1 FY -25.4 FZ 0.4
777 FX -2.1 FY 4.5 FZ -0.9
776 FX -6.8 FY -8.3 FZ 0
766 FX -4 FY -164 FZ -25
767 FX -4 FY -275 FZ -2
768 FX -167 FY 438 FZ 0
769 FX -6 FY -109 FZ -14
770 FX -6 FY -595 FZ -3
771 FX -425 FY 705 FZ 19
773 FX -47.9 FY 405.7 FZ 3.1
775 FX 0.8 FY -166.9 FZ 29.6
774 FX 0.1 FY -328.2 FZ -2.4
772 FX -47.1 FY -405.7 FZ -3.3
****
LOAD 5 LOADTYPE Wind TITLE WIND LOAD(+Z DIR)
JOINT LOAD
757 FX 0 FY 56.5 FZ 6.6
758 FX 0 FY 57.2 FZ 6.6
760 FX 0 FY -57.2 FZ 6.6
759 FX 0 FY -56.5 FZ 6.6
756 FX -0.1 FY 619.4 FZ 38.6
763 FX 0.1 FY -619.4 FZ 39
754 FX -1.3 FY 504.7 FZ 39.3
755 FX -1.9 FY 489.2 FZ 32.9
762 FX 1.9 FY -489.3 FZ 33.1
761 FX 1.3 FY -504.7 FZ 39.6
764 FX 1.7 FY 8.6 FZ 5.7
765 FX 0 FY 3.3 FZ 4.4
779 FX 0.2 FY 2.5 FZ 2.7
781 FX 0 FY 29.4 FZ 9.9
780 FX 0.5 FY -31.9 FZ 8.4
778 FX 0.5 FY 16.3 FZ 5.1
777 FX 0 FY -15.3 FZ 5.6
776 FX -1.2 FY -1 FZ 2.1
766 FX 0 FY 441 FZ 123

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767 FX 46 FY 684 FZ 138
768 FX -2 FY 465 FZ 143
769 FX 0 FY -441 FZ 123
770 FX 0 FY -683 FZ 138
771 FX -43 FY -465 FZ 143
773 FX -1 FY 166.9 FZ 29.2
775 FX 0.8 FY -166.9 FZ 29.6
774 FX -0.5 FY -170.4 FZ 29.8
772 FX 0.7 FY 170.4 FZ 30
*********************************************************************************
LOAD 6 LOADTYPE Wind  TITLE WIND LOAD(-Z DIR)
JOINT LOAD
757 FX 0 FY -56.5 FZ -6.6
758 FX 0 FY -57.2 FZ -6.6
760 FX 0 FY 57.2 FZ -6.6
759 FX 0 FY 56.5 FZ -6.6
756 FX 0.1 FY -619.4 FZ -38.6
763 FX -0.1 FY 619.4 FZ -39
754 FX 1.3 FY -504.7 FZ -39.3
755 FX 1.9 FY -489.2 FZ -32.9
762 FX -1.9 FY 489.3 FZ -33.1
761 FX -1.3 FY 504.7 FZ -39.6
764 FX -1.7 FY -8.6 FZ -5.7
765 FX 0 FY -3.3 FZ -4.4
779 FX -0.2 FY -2.5 FZ -2.7
781 FX 0 FY -29.4 FZ -9.9
780 FX -0.5 FY 31.9 FZ -8.4
778 FX -0.5 FY -16.3 FZ -5.1
777 FX 0 FY 15.3 FZ -5.6
776 FX 1.2 FY 1 FZ -2.1
766 FX 0 FY -441 FZ -123
767 FX 0 FY -683 FZ -138
768 FX -43 FY -465 FZ -143
769 FX 0 FY 441 FZ -123
770 FX 44 FY 684 FZ -136
771 FX 0 FY 465 FZ -143
773 FX 1 FY -166.9 FZ -29.2
775 FX -0.8 FY 166.9 FZ -29.6
774 FX 0.5 FY 170.4 FZ -29.8
772 FX -0.7 FY -170.4 FZ -30
*********************************************************************************
LOAD 7 LOADTYPE Temperature  TITLE TEMP PLUS
JOINT LOAD
764 FX -47.2 FY -0.7 FZ 79.8
765 FX -2.7 FY -0.9 FZ 0.7
*********************************************************************************
LOAD 8 LOADTYPE Temperature  TITLE TEMP MINUS
JOINT LOAD
764 FX 47.2 FY 0.7 FZ -79.8
765 FX 2.7 FY 0.9 FZ -0.7
*********************************************************************************
LOAD 9 LOADTYPE Dead  TITLE SDL-1
JOINT LOAD
779 FX 13.3 FY -131.6 FZ 8.6
```

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781 FX 0 FY -187.5 FZ -58.7
780 FX 7.2 FY 91.6 FZ -35
778 FX 4.7 FY -187.8 FZ -24.9
777 FX 0 FY 95 FZ -30.1
776 FX 0.7 FY -65.6 FZ 0
*******************************************************************************
LOAD 10 LOADTYPE Dead  TITLE SDL2
JOINT LOAD
779 FX -8.7 FY -55.8 FZ 0.1
781 FX 0 FY -286.7 FZ -89.8
780 FX -3.2 FY 208.3 FZ -58.9
778 FX -7.3 FY -268.3 FZ -49.7
777 FX 0 FY 191.2 FZ -64.8
776 FX -3.9 FY -70.6 FZ 0
**********LOAD COMBS FOR BASE PRESSURE*********************************************
LOAD COMB 201 DL+ SDL1+ LL
1 1.0 2 1.0 9 1.0
LOAD COMB 202 DL+ SDL2+ LL
1 1.0 2 1.0 10 1.0
LOAD COMB 203 DL+ SDL1 0.75*LL+0.75*WL+X
1 1.0 9 1.0 2 0.75 3 0.75
LOAD COMB 204 DL+SDL1 0.75*LL+0.75*WL-X
1 1.0 9 1.0 2 0.75 4 0.75
LOAD COMB 205 DL+ SDL1 0.75*LL+0.75*WL+Z
1 1.0 9 1.0 2 0.75 5 0.75
LOAD COMB 206 DL+SDL1 1 0.75*LL+0.75*WL-Z
1 1.0 9 1.0 2 0.75 6 0.75
LOAD COMB 207 DL+ SDL2 0.75*LL+0.75*WL+X
1 1.0 10 1.0 2 0.75 3 0.75
LOAD COMB 208 DL+SDL2 0.75*LL+0.75*WL-X
1 1.0 10 1.0 2 0.75 4 0.75
LOAD COMB 209 DL+ SDL2 0.75*LL+0.75*WL+Z
1 1.0 10 1.0 2 0.75 5 0.75
LOAD COMB 210 DL+SDL2 1 0.75*LL+0.75*WL-Z
1 1.0 10 1.0 2 0.75 6 0.75
LOAD COMB 211 0.6*(DL+ SDL1)+WL+X
1 0.6 9 0.6 3 1.0
LOAD COMB 212 0.6*(DL+ SDL1)+WL-X
1 0.6 9 0.6 4 1.0
LOAD COMB 213 0.6*(DL+ SDL1)+WL+Z
1 0.6 9 0.6 5 1.0
LOAD COMB 214 0.6*(DL+ SDL1)+WL-Z
1 0.6 9 0.6 6 1.0
LOAD COMB 215 0.6*(DL+ SDL2)+WL+X
1 0.6 10 0.6 3 1.0
LOAD COMB 216 0.6*(DL+ SDL2)+WL-X
1 0.6 10 0.6 4 1.0
LOAD COMB 217 0.6*(DL+ SDL2)+WL+Z
1 0.6 10 0.6 5 1.0
LOAD COMB 218 0.6*(DL+ SDL2)+WL-Z
1 0.6 10 0.6 6 1.0
LOAD COMB 219 DL+SDL1+TL+LL
1 1.0 9 1.0 7 1.0 2 1.0
LOAD COMB 220 DL+SDL2+TL+LL
```

```
1 1.0 10 1.0 7 1.0 2 1.0
LOAD COMB 221 DL+SDL1+TL+LL
1 1.0 9 1.0 8 1.0 2 1.0
LOAD COMB 222 DL+SDL2+TL+LL
1 1.0 10 1.0 8 1.0 2 1.0
**********LOAD COMBS FOR MEMB Design*****************************************
LOAD COMB 301 1.4 DL+ 1.4 SDL1
1 1.4 9 1.4
LOAD COMB 302 1.4 DL+ 1.4 SDL2
1 1.4 10 1.4
LOAD COMB 303 1.4 (DL+SDL1+ TL) +1.7 LL
1 1.4 9 1.4 7 1.4 2 1.7
LOAD COMB 304 1.4 (DL+SDL1+ TL) +1.7 LL
1 1.4 9 1.4 8 1.4 2 1.7
LOAD COMB 305 1.4 (DL+SDL2+ TL) +1.7 LL
1 1.4 10 1.4 7 1.4 2 1.7
LOAD COMB 306 1.4 (DL+SDL2+ TL) +1.7 LL
1 1.4 10 1.4 8 1.4 2 1.7
LOAD COMB 307 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+X)
1 1.2 9 1.2 2 0.5 3 1.6
LOAD COMB 308 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-X)
1 1.2 9 1.2 2 0.5 4 1.6
LOAD COMB 309 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+Z)
1 1.2 9 1.2 2 0.5 5 1.6
LOAD COMB 310 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
1 1.2 9 1.2 2 0.5 6 1.6
LOAD COMB 311 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)
1 1.2 10 1.2 2 0.5 3 1.6
LOAD COMB 312 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-X)
1 1.2 10 1.2 2 0.5 4 1.6
LOAD COMB 313 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+Z)
1 1.2 10 1.2 2 0.5 5 1.6
LOAD COMB 314 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-Z)
1 1.2 10 1.2 2 0.5 6 1.6
LOAD COMB 315 0.9*DL 0.9 SDL1 + 1.6 WL+X
1 0.9 9 0.9 3 1.6
LOAD COMB 316 0.9*DL 0.9 SDL1+ 1.6 WL-X
1 0.9 9 0.9 4 1.6
LOAD COMB 317 DL*0.9 0.9 SDL1+ 1.6 WL+Z
1 0.9 9 0.9 5 1.6
LOAD COMB 318 DL*0.9 0.9 SDL1 + 1.6 WL-Z
1 0.9 9 0.9 6 1.6
LOAD COMB 319 0.9*DL 0.9 SDL1 + 1.6 WL+X
1 0.9 10 0.9 3 1.6
LOAD COMB 320 0.9*DL 0.9 SDL1+ 1.6 WL-X
1 0.9 10 0.9 4 1.6
LOAD COMB 321 DL*0.9 0.9 SDL1+ 1.6 WL+Z
1 0.9 10 0.9 5 1.6
LOAD COMB 322 DL*0.9 0.9 SDL1 + 1.6 WL-Z
1 0.9 10 0.9 6 1.6
*******************************************************
PERFORM ANALYSIS PRINT STATICS CHECK
*******************************************************
LOAD LIST 301 TO 322
```

```
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 1 TO 12 21 TO 32 41 TO 70 81 TO 106
FYMAIN 420000 MEMB 1 TO 12 21 TO 32 41 TO 70 81 TO 106
TRACK 1 ALL
CLB 0.05 MEMB 81 TO 106
CLT 0.05 MEMB 81 TO 106
CLS 0.05 ALL
MAXMAIN 25 MEMB 1 TO 12 21 TO 32 41 TO 70
MINMAIN 16 MEMB 1 TO 12 21 TO 32 41 TO 70
MAXMAIN 16 MEMB 81 TO 106
MINMAIN 12 MEMB 81 TO 106
MINSEC 10 ALL
REINF 0 ALL
DESIGN COLUMN 1 TO 12 21 TO 32 41 TO 70
DESIGN BEAM 81 TO 106
END CONCRETE DESIGN
FINISH
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	03-09-2020	14-10-2020	31-10-2023				Date	03-09-2020	14-10-2020	31-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 5 **STAAD OUTPUT OF TOTAL FOUNDATION ARRANGEMENT**

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    SEP  2, 2020                *
*           Time=    17:32:12                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\MMGP-2020\ID FAN FDN\SDY-03.08.20\Screen Duct Support_Foundation_Final.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 28-AUG-20
    4. END JOB INFORMATION
    5. INPUT WIDTH 79
    6. UNIT METER KN
    7. JOINT COORDINATES
    8. 1 -13.39 -3.2 -10.59; 2 -12.865 -3.2 -10.59; 3 -12.34 -3.2 -10.59
    9. 4 -11.89 -3.2 -10.59; 5 -11.44 -3.2 -10.59; 6 -10.915 -3.2 -10.59
   10. 7 -10.39 -3.2 -10.59; 8 -9.89 -3.2 -10.49; 9 -9.39 -3.2 -10.49
   11. 10 -8.89 -3.2 -10.49; 11 -8.49 -3.2 -10.49; 12 -7.89 -3.2 -10.49
   12. 13 -7.39 -3.2 -10.49; 14 -6.89 -3.2 -10.49; 15 -6.39 -3.2 -10.49
   13. 16 -9.89 -3.2 -10.09; 17 -9.39 -3.2 -10.09; 18 -8.89 -3.2 -10.09
   14. 19 -8.49 -3.2 -10.09; 20 -7.89 -3.2 -10.09; 21 -7.39 -3.2 -10.09
   15. 22 -6.89 -3.2 -10.09; 23 -6.39 -3.2 -10.09; 24 -3.09 -3.2 -10.09
   16. 25 -2.59 -3.2 -10.09; 26 -2.09 -3.2 -10.09; 27 -1.59 -3.2 -10.09
   17. 28 -1.09 -3.2 -10.09; 29 -0.59 -3.2 -10.09; 30 -0.09 -3.2 -10.09
   18. 31 0.41 -3.2 -10.09; 32 0.91 -3.2 -10.09; 33 -13.39 -3.2 -10.065
   19. 34 -12.865 -3.2 -10.065; 35 -12.34 -3.2 -10.065; 36 -11.89 -3.2 -10.065
   20. 37 -11.44 -3.2 -10.065; 38 -10.915 -3.2 -10.065; 39 -10.39 -3.2 -10.065
   21. 40 -9.89 -3.2 -9.59; 41 -9.39 -3.2 -9.59; 42 -8.89 -3.2 -9.59
   22. 43 -8.49 -3.2 -9.59; 44 -7.89 -3.2 -9.59; 45 -7.39 -3.2 -9.59
   23. 46 -6.89 -3.2 -9.59; 47 -6.39 -3.2 -9.59; 48 -3.09 -3.2 -9.59
   24. 49 -2.59 -3.2 -9.59; 50 -2.09 -3.2 -9.59; 51 -1.59 -3.2 -9.59
   25. 52 -1.09 -3.2 -9.59; 53 -0.59 -3.2 -9.59; 54 -0.09 -3.2 -9.59
   26. 55 0.41 -3.2 -9.59; 56 0.91 -3.2 -9.59; 57 -13.39 -3.2 -9.54
   27. 58 -12.865 -3.2 -9.54; 59 -12.34 -3.2 -9.54; 60 -11.89 -3.2 -9.54
   28. 61 -11.44 -3.2 -9.54; 62 -10.915 -3.2 -9.54; 63 -10.39 -3.2 -9.54
   29. 64 -13.39 -3.2 -9.09; 65 -12.865 -3.2 -9.09; 66 -12.34 -3.2 -9.09
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   36. 85 -0.09 -3.2 -9.09; 86 0.41 -3.2 -9.09; 87 0.91 -3.2 -9.09
   37. 88 -13.39 -3.2 -8.64; 89 -12.865 -3.2 -8.64; 90 -12.34 -3.2 -8.64
   38. 91 -11.89 -3.2 -8.64; 92 -11.44 -3.2 -8.64; 93 -10.915 -3.2 -8.64
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39. 94 -10.39 -3.2 -8.64; 95 -9.89 -3.2 -8.59; 96 -9.39 -3.2 -8.59
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41. 100 -7.39 -3.2 -8.59; 101 -6.89 -3.2 -8.59; 102 -6.39 -3.2 -8.59
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111. 310 16.56 -3.2 -2.09; 311 17.26 -3.2 -2.09; 312 -14.47 -3.2 -1.915
112. 313 -13.82 -3.2 -1.915; 314 -13.22 -3.2 -1.915; 315 -12.72 -3.2 -1.915
113. 316 -12.22 -3.2 -1.915; 317 -11.62 -3.2 -1.915; 318 -10.97 -3.2 -1.915
114. 319 -20.25 -3.2 -1.815; 320 -19.75 -3.2 -1.815; 321 -19.25 -3.2 -1.815
115. 322 -18.75 -3.2 -1.815; 323 -18.25 -3.2 -1.815; 324 -17.75 -3.2 -1.815
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118. 331 -6.39 -3.2 -1.815; 332 -5.89 -3.2 -1.815; 333 -1.09 -3.2 -1.59
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122. 343 14.86 -3.2 -1.59; 344 15.56 -3.2 -1.59; 345 16.06 -3.2 -1.59
123. 346 16.56 -3.2 -1.59; 347 17.26 -3.2 -1.59; 348 -20.25 -3.2 -1.315
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127. 358 -12.72 -3.2 -1.315; 359 -12.22 -3.2 -1.315; 360 -11.62 -3.2 -1.315
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238. 693 21.51 -1.8 -3.565; 694 22.01 -1.8 -3.565; 695 22.51 -1.8 -3.565
239. 696 23.01 -1.8 -3.565; 697 23.51 -1.8 -3.565; 698 21.01 -1.8 -3.115
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243. 708 23.01 -1.8 -2.665; 709 23.51 -1.8 -2.665; 710 21.01 -1.8 -2.165
244. 711 21.51 -1.8 -2.165; 712 22.01 -1.8 -2.165; 713 22.51 -1.8 -2.165
245. 714 23.01 -1.8 -2.165; 715 23.51 -1.8 -2.165; 716 21.01 -1.8 -1.59
246. 717 21.51 -1.8 -1.59; 718 22.01 -1.8 -1.59; 719 22.51 -1.8 -1.59
247. 720 23.01 -1.8 -1.59; 721 23.51 -1.8 -1.59; 722 21.01 -1.8 -1.015
248. 723 21.51 -1.8 -1.015; 724 22.01 -1.8 -1.015; 725 22.51 -1.8 -1.015
249. 726 23.01 -1.8 -1.015; 727 23.51 -1.8 -1.015; 728 -7.39 -1 -9.09
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268. MEMBER INCIDENCES
269. 1 387 734; 2 734 766; 3 394 735; 4 735 767; 5 402 736; 6 736 768; 7 604 737
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283. ELEMENT INCIDENCES SHELL
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356. 535 386 387 351 350; 536 387 388 352 351; 537 388 389 353 352
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359. 544 425 426 390 389; 545 456 457 421 420; 546 457 458 422 421
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361. 550 461 462 426 425; 551 497 498 457 456; 552 498 499 458 457
362. 553 499 500 459 458; 554 500 501 460 459; 555 501 502 461 460
363. 556 502 503 462 461; 557 566 567 546 545; 558 567 568 547 546
364. 559 568 569 548 547; 560 569 570 549 548; 561 570 571 550 549
365. 562 571 572 551 550; 563 580 581 567 566; 564 581 582 568 567
366. 565 582 583 569 568; 566 583 584 570 569; 567 584 585 571 570
367. 568 585 586 572 571; 569 601 602 581 580; 570 602 603 582 581
368. 571 603 604 583 582; 572 604 605 584 583; 573 605 606 585 584
369. 574 606 607 586 585; 575 622 623 602 601; 576 623 624 603 602
370. 577 624 625 604 603; 578 625 626 605 604; 579 626 627 606 605
371. 580 627 628 607 606; 581 643 644 623 622; 582 644 645 624 623
372. 583 645 646 625 624; 584 646 647 626 625; 585 647 648 627 626
373. 586 648 649 628 627; 587 664 665 644 643; 588 665 666 645 644
374. 589 666 667 646 645; 590 667 668 647 646; 591 668 669 648 647

375. 592 669 670 649 648; 601 312 313 277 276; 602 313 314 278 277
376. 603 314 315 279 278; 604 315 316 280 279; 605 316 317 281 280
377. 606 317 318 282 281; 607 355 356 313 312; 608 356 357 314 313
378. 609 357 358 315 314; 610 358 359 316 315; 611 359 360 317 316
379. 612 360 361 318 317; 613 391 392 356 355; 614 392 393 357 356
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382. 621 436 437 394 393; 622 437 438 395 394; 623 438 439 396 395
383. 624 439 440 397 396; 625 470 471 435 434; 626 471 472 436 435
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385. 630 475 476 440 439; 631 521 522 471 470; 632 522 523 472 471
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387. 636 526 527 476 475; 637 559 560 539 538; 638 560 561 540 539
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394. 657 638 639 611 610; 658 639 640 612 611; 659 640 641 613 612
395. 660 641 642 614 613; 661 657 658 637 636; 662 658 659 638 637
396. 663 659 660 639 638; 664 660 661 640 639; 665 661 662 641 640
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398. 669 680 681 660 659; 670 681 682 661 660; 671 682 683 662 661
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400. 703 615 616 595 594; 704 629 630 616 615; 705 650 651 630 629
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402. 709 595 596 575 574; 710 616 617 596 595; 711 630 631 617 616
403. 712 651 652 631 630; 713 672 673 652 651; 714 686 687 673 672
404. 715 575 576 555 554; 716 596 597 576 575; 717 617 618 597 596
405. 718 631 632 618 617; 719 652 653 632 631; 720 673 674 653 652
406. 721 687 688 674 673; 722 576 577 556 555; 723 597 598 577 576
407. 724 618 619 598 597; 725 632 633 619 618; 726 653 654 633 632
408. 727 674 675 654 653; 728 688 689 675 674; 729 577 578 557 556
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410. 733 654 655 634 633; 734 675 676 655 654; 735 689 690 676 675
411. 736 578 579 558 557; 737 599 600 579 578; 738 620 621 600 599
412. 739 634 635 621 620; 740 655 656 635 634; 741 676 677 656 655
413. 742 690 691 677 676; 801 175 176 157 156; 802 214 215 176 175
414. 803 243 244 215 214; 804 252 253 244 243; 805 290 291 253 252
415. 806 326 327 291 290; 807 362 363 327 326; 808 398 399 363 362
416. 809 427 428 399 398; 810 463 464 428 427; 811 504 505 464 463
417. 812 176 177 158 157; 813 215 216 177 176; 814 244 245 216 215
418. 815 253 254 245 244; 816 291 292 254 253; 817 327 328 292 291
419. 818 363 364 328 327; 819 399 400 364 363; 820 428 429 400 399
420. 821 464 465 429 428; 822 505 506 465 464; 823 177 178 159 158
421. 824 216 217 178 177; 825 245 246 217 216; 826 254 255 246 245
422. 827 292 293 255 254; 828 328 329 293 292; 829 364 365 329 328
423. 830 400 401 365 364; 831 429 430 401 400; 832 465 466 430 429
424. 833 506 507 466 465; 834 178 179 160 159; 835 217 218 179 178
425. 836 246 247 218 217; 837 255 256 247 246; 838 293 294 256 255
426. 839 329 330 294 293; 840 365 366 330 329; 841 401 402 366 365
427. 842 430 431 402 401; 843 466 467 431 430; 844 507 508 467 466
428. 845 179 180 161 160; 846 218 219 180 179; 847 247 248 219 218
429. 848 256 257 248 247; 849 294 295 257 256; 850 330 331 295 294
430. 851 366 367 331 330; 852 402 403 367 366; 853 431 432 403 402

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431. 854 467 468 432 431; 855 508 509 468 467; 856 180 181 162 161
432. 857 219 220 181 180; 858 248 249 220 219; 859 257 258 249 248
433. 860 295 296 258 257; 861 331 332 296 295; 862 367 368 332 331
434. 863 403 404 368 367; 864 432 433 404 403; 865 468 469 433 432
435. 866 509 510 469 468; 867 181 182 163 162; 868 220 221 182 181
436. 869 249 250 221 220; 870 258 259 250 249; 871 182 183 164 163
437. 872 221 222 183 182; 873 250 251 222 221; 874 259 260 251 250; 901 16 17 9 8
438. 902 40 41 17 16; 903 71 72 41 40; 904 95 96 72 71; 905 119 120 96 95
439. 906 143 144 120 119; 907 17 18 10 9; 908 41 42 18 17; 909 72 73 42 41
440. 910 96 97 73 72; 911 120 121 97 96; 912 144 145 121 120; 913 18 19 11 10
441. 914 42 43 19 18; 915 73 74 43 42; 916 97 98 74 73; 917 121 122 98 97
442. 918 145 146 122 121; 919 19 20 12 11; 920 43 44 20 19; 921 74 75 44 43
443. 922 98 99 75 74; 923 122 123 99 98; 924 146 147 123 122; 925 20 21 13 12
444. 926 44 45 21 20; 927 75 76 45 44; 928 99 100 76 75; 929 123 124 100 99
445. 930 147 148 124 123; 931 21 22 14 13; 932 45 46 22 21; 933 76 77 46 45
446. 934 100 101 77 76; 935 124 125 101 100; 936 148 149 125 124; 937 22 23 15 14
447. 938 46 47 23 22; 939 77 78 47 46; 940 101 102 78 77; 941 125 126 102 101
448. 942 149 150 126 125; 943 48 49 25 24; 944 79 80 49 48; 945 103 104 80 79
449. 946 127 128 104 103; 947 49 50 26 25; 948 80 81 50 49; 949 104 105 81 80
450. 950 128 129 105 104; 951 50 51 27 26; 952 81 82 51 50; 953 105 106 82 81
451. 954 129 130 106 105; 955 51 52 28 27; 956 82 83 52 51; 957 106 107 83 82
452. 958 130 131 107 106; 959 52 53 29 28; 960 83 84 53 52; 961 107 108 84 83
453. 962 131 132 108 107; 963 53 54 30 29; 964 84 85 54 53; 965 108 109 85 84
454. 966 132 133 109 108; 967 54 55 31 30; 968 85 86 55 54; 969 109 110 86 85
455. 970 133 134 110 109; 971 55 56 32 31; 972 86 87 56 55; 973 110 111 87 86
456. 974 134 135 111 110
457. START GROUP DEFINITION
458. ELEMENT
459. _MX 203 208 213 218 223 241 242 245 TO 248 251 TO 254 257 TO 260 263 TO 266 -
460. 269 TO 272 275 TO 277 280 281 284 285 288 289 292 321 TO 330 371 TO 380 401 -
461. 402 TO 421 444 TO 449 452 TO 454 521 522 525 TO 528 531 TO 534 537 TO 540 -
462. 543 TO 546 549 TO 552 555 TO 558 561 TO 564 567 TO 570 573 TO 576 -
463. 579 TO 582 585 TO 588 591 592 601 602 605 TO 608 611 TO 614 617 TO 620 623 -
464. 624 TO 626 629 TO 632 635 TO 638 641 TO 644 647 TO 650 653 TO 656 659 TO 662 -
465. 665 TO 668 671 672 701 TO 714 729 TO 742 805 TO 811 816 TO 822 856 TO 866 -
466. 901 TO 912 937 TO 946 955 TO 962 971 TO 974
467. _MY 201 TO 210 216 TO 220 241 TO 252 265 TO 280 289 TO 292 321 325 326 329 -
468. 330 TO 331 335 336 339 TO 341 345 346 349 TO 351 355 356 360 361 365 366 370 -
469. 371 TO 372 375 376 380 TO 382 385 386 390 TO 392 395 396 399 TO 402 405 406 -
470. 409 TO 411 413 TO 415 419 TO 422 424 TO 426 430 TO 433 435 TO 437 -
471. 441 TO 444 446 TO 448 452 TO 454 521 TO 532 545 TO 568 581 TO 592 -
472. 601 TO 612 625 TO 648 661 TO 672 701 702 705 TO 709 712 TO 716 719 TO 723 -
473. 726 TO 730 733 TO 737 740 TO 742 805 TO 807 810 811 816 TO 818 821 822 827 -
474. 828 TO 829 832 833 838 TO 840 843 TO 846 849 TO 851 854 TO 857 860 TO 862 -
475. 865 TO 868 871 872 901 902 905 TO 908 911 TO 914 917 TO 920 923 TO 926 929 -
476. 930 TO 932 935 TO 938 941 TO 943 946 947 950 951 954 955 958 959 962 963 966 -
477. 967 970 971 974
478. _SHEAR 201 TO 205 241 TO 247 252 253 258 259 264 265 270 TO 277 280 289 292 -
479. 325 TO 330 335 TO 340 390 400 410 411 420 TO 422 431 TO 433 442 TO 444 453 -
480. 454 522 TO 525 527 532 533 538 539 544 545 550 552 TO 555 558 TO 561 563 -
481. 568 569 574 575 580 581 586 588 TO 591 602 TO 605 607 612 613 618 619 624 -
482. 625 630 632 TO 635 638 TO 641 643 648 649 654 655 660 661 666 668 TO 671 -
483. 702 TO 706 708 714 715 721 722 728 729 735 737 TO 741 804 TO 811 817 TO 822 -
484. 834 845 856 867 871 901 TO 907 912 913 918 919 924 925 930 931 936 937 942
485. _MX_700 601 602 605 TO 608 611 TO 614 617 TO 620 623 TO 626 629 TO 632 635 -
486. 636 TO 638 641 TO 644 647 TO 650 653 TO 656 659 TO 662 665 TO 668 671 672 -
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487. 701 TO 714 729 TO 742 805 TO 811 816 TO 822 856 TO 866
488. _MX_400 203 208 213 218 223 277 280 281 284 285 288 289 292
489. _MY_700 601 TO 612 625 TO 648 661 TO 672 701 702 705 TO 709 712 TO 716 719 -
490. 720 TO 723 726 TO 730 733 TO 737 740 TO 742 805 TO 807 810 811 816 TO 818 -
491. 821 822 827 TO 829 832 833 838 TO 840 843 TO 846 849 TO 851 854 TO 857 860 -
492. 861 TO 862 865 TO 868 871 872
493. _MY_400 201 TO 210 216 TO 220 277 TO 280 289 TO 292
494. _SHEAR_700 602 TO 605 607 612 613 618 619 624 625 630 632 TO 635 638 TO 641 -
495. 643 648 649 654 655 660 661 666 668 TO 671 702 TO 706 708 714 715 721 722 -
496. 728 729 735 737 TO 741 804 TO 811 817 TO 822 834 845 856 867 871
497. _SHEAR_400 201 TO 205 277 280 289 292
498. _MX_600 241 242 245 TO 248 251 TO 254 257 TO 260 263 TO 266 269 TO 272 275 -
499. 276 321 TO 330 371 TO 380 401 TO 421 444 TO 449 452 TO 454 521 522 -
500. 525 TO 528 531 TO 534 537 TO 540 543 TO 546 549 TO 552 555 TO 558 -
501. 561 TO 564 567 TO 570 573 TO 576 579 TO 582 585 TO 588 591 592 901 TO 912 -
502. 937 TO 946 955 TO 962 971 TO 974
503. _MY_600 241 TO 252 265 TO 276 321 325 326 329 TO 331 335 336 339 TO 341 345 -
504. 346 349 TO 351 355 356 360 361 365 366 370 TO 372 375 376 380 TO 382 385 -
505. 386 390 TO 392 395 396 399 TO 402 405 406 409 TO 411 413 TO 415 419 TO 422 -
506. 424 TO 426 430 TO 433 435 TO 437 441 TO 444 446 TO 448 452 TO 454 -
507. 521 TO 532 545 TO 568 581 TO 592 901 902 905 TO 908 911 TO 914 917 TO 920 -
508. 923 TO 926 929 TO 932 935 TO 938 941 TO 943 946 947 950 951 954 955 958 959 -
509. 962 963 966 967 970 971 974
510. _SHEAR_600 241 TO 247 252 253 258 259 264 265 270 TO 276 325 TO 330 -
511. 335 TO 340 390 400 410 411 420 TO 422 431 TO 433 442 TO 444 453 454 -
512. 522 TO 525 527 532 533 538 539 544 545 550 552 TO 555 558 TO 561 563 568 -
513. 569 574 575 580 581 586 588 TO 591 901 TO 907 912 913 918 919 924 925 930 -
514. 931 936 937 942
515. END GROUP DEFINITION
516. ELEMENT PROPERTY
517. 601 TO 672 701 TO 742 801 TO 874 THICKNESS 0.7
518. 241 TO 276 321 TO 454 521 TO 592 901 TO 974 THICKNESS 0.6
519. 201 TO 225 THICKNESS 0.4
520. 277 TO 292 THICKNESS 0.4
521. DEFINE MATERIAL START
522. ISOTROPIC CONCRETE
523. E 2.17185E+007
524. POISSON 0.17
525. DENSITY 23.5616
526. ALPHA 1E-005
527. DAMP 0.05
528. TYPE CONCRETE
529. STRENGTH FCU 35000
530. END DEFINE MATERIAL
531. MEMBER PROPERTY AMERICAN
532. 1 TO 12 PRIS YD 0.9 ZD 1.2
533. 21 TO 32 PRIS YD 0.6 ZD 0.6
534. 41 TO 48 PRIS YD 0.6 ZD 0.6
535. 51 TO 62 PRIS YD 0.7 ZD 0.5
536. 49 50 PRIS YD 0.55 ZD 0.55
537. 63 TO 70 PRIS YD 0.45 ZD 0.45
538. *
539. 82 TO 85 89 90 93 TO 95 PRIS YD 0.6 ZD 0.3
540. 102 TO 106 PRIS YD 0.4 ZD 0.3
541. 98 TO 101 PRIS YD 0.3 ZD 0.3
542. MEMBER PROPERTY
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543. 81 86 TO 88 91 92 96 97 PRIS YD 0.3 ZD 0.5
544. CONSTANTS
545. MATERIAL CONCRETE ALL
546. SUPPORTS
547. 1 TO 727 ELASTIC MAT DIRECT Y SUBGRADE 30000
548. MEMBER OFFSET
549. 63 TO 70 100 101 105 106 START 0 0 -0.225
550. 63 TO 70 100 101 105 106 END 0 0 -0.225
551. LOAD 1 LOADTYPE DEAD  TITLE DEAD LOAD
**WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
          MASTER/SLAVE OR IF UNCONNECTED JOINTS.
552. SELFWEIGHT Y -1
553. ELEMENT LOAD
554. 201 TO 225 PR GY -35.1
555. 241 TO 276 321 TO 454 521 TO 592 901 TO 974 PR GY -63.7
556. 277 TO 292 601 TO 672 701 TO 742 801 TO 874 PR GY -60.9
557. JOINT LOAD
558. **DUCT SUPPORT STRUCTURE
559. 757 FX -0.1 FY -57 FZ 0
560. 758 FX 0.1 FY -57 FZ 0
561. 760 FX 0.1 FY -3.4 FZ 0
562. 759 FX -0.1 FY -3.4 FZ 0
563. 756 FX 0.1 FY -201 FZ -0.2
564. 763 FX 0.1 FY -202 FZ 0.2
565. 754 FX -0.2 FY -139 FZ -0.2
566. 755 FX 0.1 FY -118 FZ -0.1
567. 762 FX 0.1 FY -124 FZ 0.2
568. 761 FX -0.1 FY -140 FZ 0.1
569. **** CONV 3130-003
570. 764 FX 1.5 FY -18.2 FZ -3.2
571. 765 FX -1.5 FY -25.9 FZ -5.9
572. **** TP-2
573. 779 FX 4.4 FY -72 FZ 12.3
574. 781 FX 0 FY -31.8 FZ -8.5
575. 780 FX 1.4 FY -1.6 FZ -2.9
576. 778 FX -0.8 FY -59.3 FZ 7
577. 777 FX 0 FY 2.9 FZ -6.8
578. 776 FX -1.2 FY -23.9 FZ -1.1
579. ***BAG FILTER
580. 766 FX 0 FY -598 FZ -27
581. 767 FX 0 FY -1131 FZ -51
582. 768 FX 0 FY -607 FZ -28
583. 769 FX 0 FY -582 FZ 27
584. 770 FX 0 FY -1099 FZ 51
585. 771 FX 0 FY -587 FZ 27
586. ***STAIRCASE
587. 773 FX 4 FY -57.8 FZ -0.6
588. 775 FX 0.9 FY -176.2 FZ 0.8
589. 774 FX -0.8 FY -177.2 FZ 0.6
590. 772 FX -4.1 FY -64.4 FZ -0.8
591. ********************************************************************************
592. LOAD 2 LOADTYPE LIVE  TITLE PIPE LOAD
593. JOINT LOAD
594. 757 FX 0 FY 0 FZ 0
595. 758 FX 0 FY 0 FZ 0
596. 760 FX 0 FY 0 FZ 0
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597. 759 FX 0 FY 0 FZ 0
598. 756 FX 0.4 FY 0 FZ 0
599. 763 FX 0.4 FY 0 FZ 0.1
600. 754 FX -0.4 FY 0 FZ -0.1
601. 755 FX -0.3 FY 0 FZ 0
602. 762 FX 0 FY 0 FZ 0.1
603. 761 FX -0.2 FY 0 FZ 0
604. ****CONV 3130-003
605. 764 FX -4.5 FY -33.9 FZ -5.1
606. 765 FX 0.1 FY -43.2 FZ -18
607. 779 FX 2.8 FY -64.1 FZ 5.6
608. 781 FX 0 FY -27.5 FZ -6.3
609. 780 FX 4.7 FY -26.6 FZ 0.9
610. 778 FX -2.9 FY -45.7 FZ 2.1
611. 777 FX 0 FY -26.7 FZ 1.5
612. 776 FX -4.5 FY -57.7 FZ -3.8
613. 766 FX 0.1 FY 0.1 FZ 0.1
614. 767 FX 0.1 FY 0.1 FZ 0.1
615. 768 FX 0.1 FY 0.1 FZ 0.1
616. 769 FX 0.1 FY 0.1 FZ 0.1
617. 770 FX 0 FY 0 FZ 0
618. 771 FX 0 FY 0 FZ 0
619. 773 FX 9.9 FY -114.7 FZ -0.1
620. 775 FX 0.5 FY -102 FZ 0.2
621. 774 FX -0.5 FY -115.8 FZ 0.1
622. 772 FX -10 FY -132.5 FZ -0.2
623. ******************************************************************************
624. LOAD 3 LOADTYPE WIND  TITLE WIND LOAD (+X DIR)
625. JOINT LOAD
626. 757 FX 6 FY 34.4 FZ -0.7
627. 758 FX 5.7 FY -34.4 FZ 0.7
628. 760 FX 4 FY -36.1 FZ 0.8
629. 759 FX 4.2 FY 36.1 FZ -0.8
630. 756 FX 4.3 FY -16.1 FZ -0.1
631. 763 FX 4.3 FY -11.6 FZ -0.1
632. 754 FX 21.9 FY 303.6 FZ 0.8
633. 755 FX 22.8 FY -287.5 FZ -0.7
634. 762 FX 25.1 FY -297.3 FZ -0.6
635. 761 FX 24.1 FY 308.9 FZ 0.7
636. 779 FX 16.1 FY -24.2 FZ 2.1
637. 781 FX 2 FY 7.6 FZ -0.9
638. 780 FX 4.9 FY -12.6 FZ -1.6
639. 778 FX 16.1 FY 25.4 FZ -0.4
640. 777 FX 2.1 FY -4.5 FZ 0.9
641. 776 FX 6.8 FY 8.3 FZ 0
642. 766 FX 4 FY 164 FZ 26
643. 767 FX 171 FY 278 FZ 1
644. 768 FX 3 FY -442 FZ -28
645. 769 FX 7 FY 105 FZ 15
646. 770 FX 426 FY 598 FZ 3
647. 771 FX 2 FY -704 FZ -18
648. 773 FX 47.9 FY -405.7 FZ -3.1
649. 775 FX 84 FY -328.2 FZ -2.5
650. 774 FX -0.1 FY 328.2 FZ 2.4
651. 772 FX 47.1 FY 405.7 FZ 3.3
652. ******************************************************************************
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653. LOAD 4 LOADTYPE WIND  TITLE WIND LOAD (-X DIR)
654. JOINT LOAD
655. 757 FX -6 FY -34.4 FZ 0.7
656. 758 FX -5.7 FY 34.4 FZ -0.7
657. 760 FX -4 FY 36.1 FZ -0.8
658. 759 FX -4.2 FY -36.1 FZ 0.8
659. 756 FX -4.3 FY 16.1 FZ 0.1
660. 763 FX -4.3 FY 11.6 FZ 0.1
661. 754 FX -21.9 FY -303.6 FZ -0.8
662. 755 FX -22.8 FY 287.5 FZ 0.7
663. 762 FX -25.1 FY 297.3 FZ 0.6
664. 761 FX -24.1 FY -308.9 FZ -0.7
665. 779 FX -16.1 FY 24.2 FZ -2.1
666. 781 FX -2 FY -7.6 FZ 0.9
667. 780 FX -4.9 FY 12.6 FZ 1.6
668. 778 FX -16.1 FY -25.4 FZ 0.4
669. 777 FX -2.1 FY 4.5 FZ -0.9
670. 776 FX -6.8 FY -8.3 FZ 0
671. 766 FX -4 FY -164 FZ -25
672. 767 FX -4 FY -275 FZ -2
673. 768 FX -167 FY 438 FZ 0
674. 769 FX -6 FY -109 FZ -14
675. 770 FX -6 FY -595 FZ -3
676. 771 FX -425 FY 705 FZ 19
677. 773 FX -47.9 FY 405.7 FZ 3.1
678. 775 FX 0.8 FY -166.9 FZ 29.6
679. 774 FX 0.1 FY -328.2 FZ -2.4
680. 772 FX -47.1 FY -405.7 FZ -3.3
681. ******************************************************************************
682. LOAD 5 LOADTYPE WIND  TITLE WIND LOAD(+Z DIR)
683. JOINT LOAD
684. 757 FX 0 FY 56.5 FZ 6.6
685. 758 FX 0 FY 57.2 FZ 6.6
686. 760 FX 0 FY -57.2 FZ 6.6
687. 759 FX 0 FY -56.5 FZ 6.6
688. 756 FX -0.1 FY 619.4 FZ 38.6
689. 763 FX 0.1 FY -619.4 FZ 39
690. 754 FX -1.3 FY 504.7 FZ 39.3
691. 755 FX -1.9 FY 489.2 FZ 32.9
692. 762 FX 1.9 FY -489.3 FZ 33.1
693. 761 FX 1.3 FY -504.7 FZ 39.6
694. 764 FX 1.7 FY 8.6 FZ 5.7
695. 765 FX 0 FY 3.3 FZ 4.4
696. 779 FX 0.2 FY 2.5 FZ 2.7
697. 781 FX 0 FY 29.4 FZ 9.9
698. 780 FX 0.5 FY -31.9 FZ 8.4
699. 778 FX 0.5 FY 16.3 FZ 5.1
700. 777 FX 0 FY -15.3 FZ 5.6
701. 776 FX -1.2 FY -1 FZ 2.1
702. 766 FX 0 FY 441 FZ 123
703. 767 FX 46 FY 684 FZ 138
704. 768 FX -2 FY 465 FZ 143
705. 769 FX 0 FY -441 FZ 123
706. 770 FX 0 FY -683 FZ 138
707. 771 FX -43 FY -465 FZ 143
708. 773 FX -1 FY 166.9 FZ 29.2
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775 FX 0.8 FY -166.9 FZ 29.6
774 FX -0.5 FY -170.4 FZ 29.8
772 FX 0.7 FY 170.4 FZ 30
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LOAD 6 LOADTYPE WIND  TITLE WIND LOAD(-Z DIR)
JOINT LOAD
757 FX 0 FY -56.5 FZ -6.6
758 FX 0 FY -57.2 FZ -6.6
760 FX 0 FY 57.2 FZ -6.6
759 FX 0 FY 56.5 FZ -6.6
756 FX 0.1 FY -619.4 FZ -38.6
763 FX -0.1 FY 619.4 FZ -39
754 FX 1.3 FY -504.7 FZ -39.3
755 FX 1.9 FY -489.2 FZ -32.9
762 FX -1.9 FY 489.3 FZ -33.1
761 FX -1.3 FY 504.7 FZ -39.6
764 FX -1.7 FY -8.6 FZ -5.7
765 FX 0 FY -3.3 FZ -4.4
779 FX -0.2 FY -2.5 FZ -2.7
781 FX 0 FY -29.4 FZ -9.9
780 FX -0.5 FY 31.9 FZ -8.4
778 FX -0.5 FY -16.3 FZ -5.1
777 FX 0 FY 15.3 FZ -5.6
776 FX 1.2 FY 1 FZ -2.1
766 FX 0 FY -441 FZ -123
767 FX 0 FY -683 FZ -138
768 FX -43 FY -465 FZ -143
769 FX 0 FY 441 FZ -123
770 FX 44 FY 684 FZ -136
771 FX 0 FY 465 FZ -143
773 FX 1 FY -166.9 FZ -29.2
775 FX -0.8 FY 166.9 FZ -29.6
774 FX 0.5 FY 170.4 FZ -29.8
772 FX -0.7 FY -170.4 FZ -30
********************************************************************************
LOAD 7 LOADTYPE TEMPERATURE  TITLE TEMP PLUS
JOINT LOAD
764 FX -47.2 FY -0.7 FZ 79.8
765 FX -2.7 FY -0.9 FZ 0.7
********************************************************************************
LOAD 8 LOADTYPE TEMPERATURE  TITLE TEMP MINUS
JOINT LOAD
764 FX 47.2 FY 0.7 FZ -79.8
765 FX 2.7 FY 0.9 FZ -0.7
********************************************************************************
LOAD 9 LOADTYPE DEAD  TITLE SDL-1
JOINT LOAD
779 FX 13.3 FY -131.6 FZ 8.6
781 FX 0 FY -187.5 FZ -58.7
780 FX 7.2 FY 91.6 FZ -35
778 FX 4.7 FY -187.8 FZ -24.9
777 FX 0 FY 95 FZ -30.1
776 FX 0.7 FY -65.6 FZ 0
********************************************************************************
LOAD 10 LOADTYPE DEAD  TITLE SDL2
JOINT LOAD
```

765. 779 FX -8.7 FY -55.8 FZ 0.1
766. 781 FX 0 FY -286.7 FZ -89.8
767. 780 FX -3.2 FY 208.3 FZ -58.9
768. 778 FX -7.3 FY -268.3 FZ -49.7
769. 777 FX 0 FY 191.2 FZ -64.8
770. 776 FX -3.9 FY -70.6 FZ 0
771. **********LOAD COMBS FOR BASE PRESSURE**********
772. LOAD COMB 201 DL+ SDL1+ LL
773. 1 1.0 2 1.0 9 1.0
774. LOAD COMB 202 DL+ SDL2+ LL
775. 1 1.0 2 1.0 10 1.0
776. LOAD COMB 203 DL+ SDL1 0.75*LL+0.75*WL+X
777. 1 1.0 9 1.0 2 0.75 3 0.75
778. LOAD COMB 204 DL+SDL1 0.75*LL+0.75*WL-X
779. 1 1.0 9 1.0 2 0.75 4 0.75
780. LOAD COMB 205 DL+ SDL1 0.75*LL+0.75*WL+Z
781. 1 1.0 9 1.0 2 0.75 5 0.75
782. LOAD COMB 206 DL+SDL1 1 0.75*LL+0.75*WL-Z
783. 1 1.0 9 1.0 2 0.75 6 0.75
784. LOAD COMB 207 DL+ SDL2 0.75*LL+0.75*WL+X
785. 1 1.0 10 1.0 2 0.75 3 0.75
786. LOAD COMB 208 DL+SDL2 0.75*LL+0.75*WL-X
787. 1 1.0 10 1.0 2 0.75 4 0.75
788. LOAD COMB 209 DL+ SDL2 0.75*LL+0.75*WL+Z
789. 1 1.0 10 1.0 2 0.75 5 0.75
790. LOAD COMB 210 DL+SDL2 1 0.75*LL+0.75*WL-Z
791. 1 1.0 10 1.0 2 0.75 6 0.75
792. LOAD COMB 211 0.6*(DL+ SDL1)+WL+X
793. 1 0.6 9 0.6 3 1.0
794. LOAD COMB 212 0.6*(DL+ SDL1)+WL-X
795. 1 0.6 9 0.6 4 1.0
796. LOAD COMB 213 0.6*(DL+ SDL1)+WL+Z
797. 1 0.6 9 0.6 5 1.0
798. LOAD COMB 214 0.6*(DL+ SDL1)+WL-Z
799. 1 0.6 9 0.6 6 1.0
800. LOAD COMB 215 0.6*(DL+ SDL2)+WL+X
801. 1 0.6 10 0.6 3 1.0
802. LOAD COMB 216 0.6*(DL+ SDL2)+WL-X
803. 1 0.6 10 0.6 4 1.0
804. LOAD COMB 217 0.6*(DL+ SDL2)+WL+Z
805. 1 0.6 10 0.6 5 1.0
806. LOAD COMB 218 0.6*(DL+ SDL2)+WL-Z
807. 1 0.6 10 0.6 6 1.0
808. LOAD COMB 219 DL+SDL1+TL+LL
809. 1 1.0 9 1.0 7 1.0 2 1.0
810. LOAD COMB 220 DL+SDL2+TL+LL
811. 1 1.0 10 1.0 7 1.0 2 1.0
812. LOAD COMB 221 DL+SDL1+TL+LL
813. 1 1.0 9 1.0 8 1.0 2 1.0
814. LOAD COMB 222 DL+SDL2+TL+LL
815. 1 1.0 10 1.0 8 1.0 2 1.0
816. **********LOAD COMBS FOR MEMB DESIGN**********
817. LOAD COMB 301 1.4 DL+ 1.4 SDL1
818. 1 1.4 9 1.4
819. LOAD COMB 302 1.4 DL+ 1.4 SDL2
820. 1 1.4 10 1.4

```
821. LOAD COMB 303 1.4 (DL+SDL1+ TL) +1.7 LL
822. 1 1.4 9 1.4 7 1.4 2 1.7
823. LOAD COMB 304 1.4 (DL+SDL1+ TL) +1.7 LL
824. 1 1.4 9 1.4 8 1.4 2 1.7
825. LOAD COMB 305 1.4 (DL+SDL2+ TL) +1.7 LL
826. 1 1.4 10 1.4 7 1.4 2 1.7
827. LOAD COMB 306 1.4 (DL+SDL2+ TL) +1.7 LL
828. 1 1.4 10 1.4 8 1.4 2 1.7
829. LOAD COMB 307 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+X)
830. 1 1.2 9 1.2 2 0.5 3 1.6
831. LOAD COMB 308 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-X)
832. 1 1.2 9 1.2 2 0.5 4 1.6
833. LOAD COMB 309 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+Z)
834. 1 1.2 9 1.2 2 0.5 5 1.6
835. LOAD COMB 310 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
836. 1 1.2 9 1.2 2 0.5 6 1.6
837. LOAD COMB 311 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)
838. 1 1.2 10 1.2 2 0.5 3 1.6
839. LOAD COMB 312 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-X)
840. 1 1.2 10 1.2 2 0.5 4 1.6
841. LOAD COMB 313 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+Z)
842. 1 1.2 10 1.2 2 0.5 5 1.6
843. LOAD COMB 314 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-Z)
844. 1 1.2 10 1.2 2 0.5 6 1.6
845. LOAD COMB 315 0.9*DL 0.9 SDL1 + 1.6 WL+X
846. 1 0.9 9 0.9 3 1.6
847. LOAD COMB 316 0.9*DL 0.9 SDL1+ 1.6 WL-X
848. 1 0.9 9 0.9 4 1.6
849. LOAD COMB 317 DL*0.9 0.9 SDL1+ 1.6 WL+Z
850. 1 0.9 9 0.9 5 1.6
851. LOAD COMB 318 DL*0.9 0.9 SDL1 + 1.6 WL-Z
852. 1 0.9 9 0.9 6 1.6
853. LOAD COMB 319 0.9*DL 0.9 SDL1 + 1.6 WL+X
854. 1 0.9 10 0.9 3 1.6
855. LOAD COMB 320 0.9*DL 0.9 SDL1+ 1.6 WL-X
856. 1 0.9 10 0.9 4 1.6
857. LOAD COMB 321 DL*0.9 0.9 SDL1+ 1.6 WL+Z
858. 1 0.9 10 0.9 5 1.6
859. LOAD COMB 322 DL*0.9 0.9 SDL1 + 1.6 WL-Z
860. 1 0.9 10 0.9 6 1.6
861. ******************************************************
862. PERFORM ANALYSIS PRINT STATICS CHECK
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**781**	**NUMBER OF MEMBERS**	**80**
NUMBER OF PLATES	**545**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**727**

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 10, TOTAL DEGREES OF FREEDOM = 2505
TOTAL LOAD COMBINATION CASES = 44 SO FAR.

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 1
LOADTYPE DEAD TITLE DEAD LOAD

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.654780668E+01
Y = -0.169114808E+01
Z = -0.355723856E+00

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 3.90
SUMMATION FORCE-Y = -18939.47
SUMMATION FORCE-Z = -10.10

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -6741.15 MY= -0.32 MZ= 124007.97

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = -3.90
SUMMATION FORCE-Y = 18939.47
SUMMATION FORCE-Z = 10.10

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 6741.15 MY= 0.31 MZ= -124007.97

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 1)

	MAXIMUMS	AT NODE
X =	2.17458E-01	760
Y =	-4.97657E-01	664
Z =	-1.90472E-01	764
RX=	1.05449E-03	546
RY=	-1.58419E-04	778
RZ=	-1.10829E-03	722

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 2
LOADTYPE LIVE TITLE PIPE LOAD

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.617342408E+01
Y = 0.864113965E+00
Z = -0.595275316E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = -4.10
SUMMATION FORCE-Y = -790.00
SUMMATION FORCE-Z = -22.60

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -4710.42 MY= 300.84 MZ= 4878.26

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 4.10
SUMMATION FORCE-Y = 790.00
SUMMATION FORCE-Z = 22.60

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 4710.42 MY= -300.84 MZ= -4878.26

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 2)

	MAXIMUMS	AT NODE
X =	-5.06387E-02	764
Y =	-5.35221E-02	8
Z =	-1.96742E-01	765
RX=	-7.74946E-04	765
RY=	-2.62270E-05	777
RZ=	1.97830E-04	764

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
LOADTYPE WIND TITLE WIND LOAD (+X DIR)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.922718183E+01
Y = 0.732676928E+00
Z = 0.110499874E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 962.30
SUMMATION FORCE-Y = -1.00
SUMMATION FORCE-Z = -0.80

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 177.70 MY= 1577.95 MZ= -13034.14

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = -962.30
SUMMATION FORCE-Y = 1.00
SUMMATION FORCE-Z = 0.80

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -177.70 MY= -1577.95 MZ= 13034.14

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 3)

	MAXIMUMS	AT NODE
X =	5.75766E-01	770
Y =	1.51978E-01	678
Z =	5.42101E-02	766
RX=	-6.87806E-04	151
RY=	8.02351E-05	773
RZ=	-1.93389E-03	770

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 4
LOADTYPE WIND TITLE WIND LOAD (-X DIR)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.528671430E+01
Y = 0.721306337E+00
Z = 0.162691078E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = -876.50
SUMMATION FORCE-Y = -495.10
SUMMATION FORCE-Z = 1.90

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2199.93 MY= -1871.97 MZ= 17184.26

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 876.50
SUMMATION FORCE-Y = 495.10
SUMMATION FORCE-Z = -1.90

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 2199.93 MY= 1871.97 MZ= -17184.26

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 4)

	MAXIMUMS	AT NODE
X =	-6.23678E-01	771
Y =	-1.49281E-01	678
Z =	1.23561E-01	775
RX=	5.77466E-04	151
RY=	-4.47926E-05	773
RZ=	2.08380E-03	771

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 5
LOADTYPE WIND TITLE WIND LOAD(+Z DIR)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.760695929E+01
Y = 0.683081847E+00
Z = 0.830735494E+00

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = 2.70
SUMMATION FORCE-Y = 12.80
SUMMATION FORCE-Z = 1219.40

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 17796.51 MY= 8964.73 MZ= 70.12

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = -2.70
SUMMATION FORCE-Y = -12.80
SUMMATION FORCE-Z = -1219.40

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -17796.51 MY= -8964.73 MZ= -70.12

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 5)

	MAXIMUMS	AT NODE
X =	-1.06942E-01	771
Y =	-2.85619E-01	664
Z =	4.78585E-01	769
RX=	1.27935E-03	769
RY=	9.22890E-05	774
RZ=	-5.61277E-04	289

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 6
LOADTYPE WIND TITLE WIND LOAD(-Z DIR)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.759855936E+01
Y = 0.682971911E+00
Z = 0.821117853E+00

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = -0.70
SUMMATION FORCE-Y = -10.80
SUMMATION FORCE-Z = -1217.40

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -17800.88 MY= -8933.42 MZ= -97.06

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = 0.70
SUMMATION FORCE-Y = 10.80
SUMMATION FORCE-Z = 1217.40

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 17800.88 MY= 8933.42 MZ= 97.06

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 6)

	MAXIMUMS	AT NODE
X =	1.05175E-01	770
Y =	2.87562E-01	664
Z =	-4.78423E-01	769
RX=	-1.27891E-03	769
RY=	-9.24612E-05	752
RZ=	5.54682E-04	289

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 7
LOADTYPE TEMPERATURE TITLE TEMP PLUS

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.175679348E+01
Y = 0.349999998E+00
Z = -0.899999984E-01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.971312474E+01
Y = 0.349999998E+00
Z = -0.899999984E-01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.104608692E+01
Y = 0.349999998E+00
Z = -0.899999984E-01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = -49.90
SUMMATION FORCE-Y = -1.60
SUMMATION FORCE-Z = 80.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 28.03 MY= -79.72 MZ= 1.92

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = 49.90
SUMMATION FORCE-Y = 1.60
SUMMATION FORCE-Z = -80.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -28.03 MY= 79.72 MZ= -1.92

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 7)

	MAXIMUMS	AT NODE
X =	-5.03901E-01	764
Y =	-1.47977E-02	514
Z =	8.64573E-01	764
RX=	3.41404E-03	764
RY=	2.58697E-06	777
RZ=	1.99828E-03	764

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 8
LOADTYPE TEMPERATURE TITLE TEMP MINUS

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.175679348E+01
Y = 0.349999998E+00
Z = -0.899999984E-01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.971312474E+01
Y = 0.349999998E+00
Z = -0.899999984E-01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.104608692E+01
Y = 0.349999998E+00
Z = -0.899999984E-01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 8)
SUMMATION FORCE-X = 49.90
SUMMATION FORCE-Y = 1.60
SUMMATION FORCE-Z = -80.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -28.03 MY= 79.72 MZ= -1.92

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 8)
SUMMATION FORCE-X = -49.90
SUMMATION FORCE-Y = -1.60
SUMMATION FORCE-Z = 80.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 28.03 MY= -79.72 MZ= 1.92

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 8)

	MAXIMUMS	AT NODE
X =	5.03901E-01	764
Y =	1.47977E-02	514
Z =	-8.64573E-01	764
RX=	-3.41404E-03	764
RY=	-2.58697E-06	777
RZ=	-1.99828E-03	764

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 9
LOADTYPE DEAD TITLE SDL-1

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.231664086E+01
Y = 0.105000004E+01
Z = -0.441799228E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.456401859E+01
Y = 0.105000004E+01
Z = -0.543625262E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 9)
SUMMATION FORCE-X = 25.90
SUMMATION FORCE-Y = -385.90
SUMMATION FORCE-Z = -140.10

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2309.68 MY= -753.84 MZ= 621.49

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 9)
SUMMATION FORCE-X = -25.90
SUMMATION FORCE-Y = 385.90
SUMMATION FORCE-Z = 140.10

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 2309.68 MY= 753.85 MZ= -621.49

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 9)

	MAXIMUMS	AT NODE
X =	8.75560E-02	779
Y =	-8.12553E-02	32
Z =	-5.99126E-01	781
RX=	-2.05567E-03	781
RY=	1.87729E-04	732
RZ=	-3.12531E-04	779

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 10
LOADTYPE DEAD TITLE SDL2

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.233372302E+01
Y = 0.105000004E+01
Z = -0.605558446E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.402580382E+01
Y = 0.105000004E+01
Z = -0.570111749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 10)
SUMMATION FORCE-X = -23.10
SUMMATION FORCE-Y = -281.90
SUMMATION FORCE-Z = -263.10

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2041.90 MY= -919.30 MZ= 181.53

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 10)
SUMMATION FORCE-X = 23.10
SUMMATION FORCE-Y = 281.90
SUMMATION FORCE-Z = 263.10

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 2041.90 MY= 919.31 MZ= -181.53

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 10)

	MAXIMUMS	AT NODE
X =	5.48042E-02	777
Y =	-1.37453E-01	32
Z =	-9.40298E-01	781
RX=	-3.21104E-03	781
RY=	2.69364E-04	781
RZ=	-2.23236E-04	24

********** END OF DATA FROM INTERNAL STORAGE **********

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863. ************************************************************
864. LOAD LIST 301 TO 322
865. START CONCRETE DESIGN
```

```
866. CODE ACI 2008
867. FC 35000 MEMB 1 TO 12 21 TO 32 41 TO 70 81 TO 106
868. FYMAIN 420000 MEMB 1 TO 12 21 TO 32 41 TO 70 81 TO 106
869. TRACK 1 ALL
870. CLB 0.05 MEMB 81 TO 106
871. CLT 0.05 MEMB 81 TO 106
872. CLS 0.05 ALL
873. MAXMAIN 25 MEMB 1 TO 12 21 TO 32 41 TO 70
874. MINMAIN 16 MEMB 1 TO 12 21 TO 32 41 TO 70
875. MAXMAIN 16 MEMB 81 TO 106
876. MINMAIN 12 MEMB 81 TO 106
877. MINSEC 10 ALL
878. REINF 0 ALL
879. DESIGN COLUMN 1 TO 12 21 TO 32 41 TO 70
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COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1200.0 X 900.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10800.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
56 - 16 MM	1.043	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
36524.05	29219.24	14172.81	4775.46	336.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
1901.10	-4729.03	1416.09	31.90	0.00901

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
36524.05	29219.24	14456.57	6448.43	446.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
2581.05	-4729.03	1416.09	87.44	0.02470

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COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1200.0 X 900.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10800.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
56 - 16 MM	1.043	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
36524.05  29219.24  14172.81   4775.46     336.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
 1901.10  -4729.03   1288.00      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
36524.05  29219.24  14456.57   6448.43     446.1
    M0       P-tens.   Des.Pn    Des.Mn     e/h
 2581.05  -4729.03   1288.00      0.00    0.00000
 ---------------------------------------------------------
```

==========

COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  RECT SIZE -1200.0 X 900.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED = 10800.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

 56 - 16 MM               1.043      301      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 256.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
36524.05  29219.24  14172.81   4775.46     336.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
 1901.10  -4729.03   2583.99      7.36    0.00114
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
36524.05  29219.24  14456.57   6448.43     446.1
    M0       P-tens.   Des.Pn    Des.Mn     e/h
 2581.05  -4729.03   2583.99    136.56    0.02114
 ---------------------------------------------------------
```

==========

COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1200.0 X 900.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10800.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
56 - 16 MM	1.043	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
36524.05	29219.24	14172.81	4775.46	336.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
1901.10	-4729.03	2436.00	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
36524.05	29219.24	14456.57	6448.43	446.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
2581.05	-4729.03	2436.00	0.00	0.00000

==========

COLUMN NO. 5 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1200.0 X 900.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10800.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
56 - 16 MM	1.043	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
36524.05	29219.24	14172.81	4775.46	336.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
1901.10	-4729.03	1439.88	24.31	0.00675

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
36524.05  29219.24  14456.57   6448.43     446.1
    M0       P-tens.   Des.Pn    Des.Mn     e/h
 2581.05  -4729.03   1439.88     55.81    0.01550
 ---------------------------------------------------------
```

==========

COLUMN NO. 6 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  RECT SIZE -1200.0 X 900.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED = 10800.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

 56 - 16 MM               1.043      301      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 256.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
36524.05  29219.24  14172.81   4775.46     336.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
 1901.10  -4729.03   1307.38      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
36524.05  29219.24  14456.57   6448.43     446.1
    M0       P-tens.   Des.Pn    Des.Mn     e/h
 2581.05  -4729.03   1307.38      0.00    0.00000
 ---------------------------------------------------------
```

==========

COLUMN NO. 7 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  RECT SIZE -1200.0 X 900.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED = 10800.0  SQ. MM
```

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

56 - 16 MM 1.043 301 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
36524.05	29219.24	14172.81	4775.46	336.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
1901.10	-4729.03	1408.03	41.46	0.01178

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
36524.05	29219.24	14456.57	6448.43	446.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
2581.05	-4729.03	1408.03	8.18	0.00232

==========

COLUMN NO. 8 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1200.0 X 900.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10800.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

56 - 16 MM 1.043 301 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
36524.05	29219.24	14172.81	4775.46	336.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
1901.10	-4729.03	1253.54	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
36524.05	29219.24	14456.57	6448.43	446.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
2581.05	-4729.03	1253.54	0.00	0.00000

==========

COLUMN NO. 9 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1200.0 X 900.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10800.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
56 - 16 MM	1.043	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
36524.05	29219.24	14172.81	4775.46	336.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
1901.10	-4729.03	2530.12	25.43	0.00402

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
36524.05	29219.24	14456.57	6448.43	446.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
2581.05	-4729.03	2530.12	86.92	0.01374

==========

COLUMN NO. 10 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1200.0 X 900.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10800.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
56 - 16 MM	1.043	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
36524.05  29219.24  14172.81   4775.46     336.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
 1901.10  -4729.03   2367.08      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
36524.05  29219.24  14456.57   6448.43     446.1
    M0       P-tens.   Des.Pn    Des.Mn     e/h
 2581.05  -4729.03   2367.08      0.00    0.00000
 ---------------------------------------------------------
```

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COLUMN NO. 11 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  RECT SIZE -1200.0 X 900.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED = 10800.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

 56 - 16 MM               1.043      301      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 256.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
36524.05  29219.24  14172.81   4775.46     336.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
 1901.10  -4729.03   1402.67     16.25    0.00464
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
36524.05  29219.24  14456.57   6448.43     446.1
    M0       P-tens.   Des.Pn    Des.Mn     e/h
 2581.05  -4729.03   1402.67     26.19    0.00747
 ---------------------------------------------------------
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COLUMN NO. 12 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1200.0 X 900.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10800.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
56 - 16 MM	1.043	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
36524.05	29219.24	14172.81	4775.46	336.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
1901.10	-4729.03	1264.31	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
36524.05	29219.24	14456.57	6448.43	446.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
2581.05	-4729.03	1264.31	0.00	0.00000

==========

COLUMN NO. 21 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	278.91	22.35	0.03643

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    278.91     48.92    0.07973
 ---------------------------------------------------------
```

==========

COLUMN NO. 22 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

 12 - 20 MM               1.047      301      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    192.77      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    192.77      0.00    0.00000
 ---------------------------------------------------------
```

==========

COLUMN NO. 23 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM
```

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 301 END 0.900
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	-48.38	19.84	-0.18637

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	-48.38	85.21	-0.80058

==========

COLUMN NO. 24 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 301 END 0.900
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	-152.29	0.00	-0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	-152.29	0.00	-0.00000

==========

COLUMN NO. 25 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	552.09	52.96	0.04361

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	552.09	61.68	0.05078

==========

COLUMN NO. 26 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    532.22      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    532.22      0.00    0.00000
 ---------------------------------------------------------
```

==========

COLUMN NO. 27 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        AREA OF STEEL REQUIRED =  5212.8  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

 28 - 16 MM               1.564      312      STA     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 256.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12907.03  10325.62   4532.16   1159.19     255.8
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  594.93  -2364.51    281.66    161.03    0.25987
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12907.03  10325.62   4532.16   1159.19     255.8
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  594.93  -2364.51    281.66    520.51    0.84001
 ---------------------------------------------------------
```

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COLUMN NO. 28 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	438.52	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	438.52	0.00	0.00000

==========

COLUMN NO. 29 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 4507.2 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 16 MM	1.340	314	STA	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12593.17	10074.53	4523.92	1106.53	244.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
518.62	-2026.73	-269.01	25.53	-0.04314

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12593.17  10074.53   4523.92   1106.53     244.6
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  518.62  -2026.73   -269.01    411.79   -0.69579
 ---------------------------------------------------------
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COLUMN NO. 30 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

 12 - 20 MM               1.047      301      END     0.900
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37   -140.00      0.00   -0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37   -140.00      0.00   -0.00000
 ---------------------------------------------------------
```

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COLUMN NO. 31 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        AREA OF STEEL REQUIRED =  5994.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------
```

20 - 20 MM 1.745 314 STA 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13162.02	10529.62	4504.02	1201.07	266.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
657.69	-2638.95	778.92	130.47	0.07614

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13162.02	10529.62	4504.02	1201.07	266.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
657.69	-2638.95	778.92	720.70	0.42057

==========

COLUMN NO. 32 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 301 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	472.34	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	472.34	0.00	0.00000

==========

COLUMN NO. 41 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	198.44	1.47	0.00247

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	198.44	13.48	0.02264

==========

COLUMN NO. 42 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ------------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    138.71      0.00    0.00000
 ------------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ------------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    138.71      0.00    0.00000
 ------------------------------------------------------------
```

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COLUMN NO. 43 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        AREA OF STEEL REQUIRED =  4129.2  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ------------------------------------------------------------

 24 - 16 MM               1.340      320      STA     0.900
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 256.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ------------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12593.17  10074.53   4523.92   1106.53     244.6
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  518.62  -2026.73   -635.04    236.35   -0.12406
 ------------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ------------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12593.17  10074.53   4523.92   1106.53     244.6
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  518.62  -2026.73   -635.04     87.49   -0.04592
 ------------------------------------------------------------
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COLUMN NO. 44 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 3625.2 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	317	END	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	-238.91	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	-238.91	0.00	-0.00000

==========

COLUMN NO. 45 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	435.50	11.33	0.00867

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    435.50     11.34    0.00868
 ---------------------------------------------------------
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COLUMN NO. 46 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

 12 - 20 MM               1.047      301      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    381.66      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    381.66      0.00    0.00000
 ---------------------------------------------------------
```

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COLUMN NO. 47 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        AREA OF STEEL REQUIRED =  3675.6  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------
```

12 - 20 MM 1.047 310 STA 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	21.67	1.12	0.01724

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	21.67	406.43	6.25063

==========

COLUMN NO. 48 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 301 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	379.51	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	379.51	0.00	0.00000

==========

COLUMN NO. 49 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
AREA OF STEEL REQUIRED = 9885.7 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 25 MM	3.895	304	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13596.86	10877.49	3666.10	1289.32	351.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
1009.11	-4947.97	180.85	330.59	0.51492

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13596.86	10877.49	3666.10	1289.32	351.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
1009.11	-4947.97	180.85	681.98	1.06224

==========

COLUMN NO. 50 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.063	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
10254.82   8203.86   3752.05    797.52     212.6
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  320.56  -1351.15     55.78      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
10254.82   8203.86   3752.05    797.52     212.6
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  320.56  -1351.15     55.78      0.00    0.00000
 ---------------------------------------------------------
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COLUMN NO. 51 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  RECT SIZE - 500.0 X 700.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3500.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

 12 - 20 MM               1.077      301      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4455.00   1204.46     270.4
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  485.93  -1583.37    311.85      0.98    0.00114
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4260.51    834.62     195.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  337.39  -1583.37    311.85      1.67    0.00195
 ---------------------------------------------------------
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COLUMN NO. 52 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	299.38	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	299.38	0.00	0.00000

==========

COLUMN NO. 53 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	332.53	3.77	0.00412

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	332.53	11.48	0.01255

==========

COLUMN NO. 54 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	301.54	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	301.54	0.00	0.00000

==========

COLUMN NO. 55 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.077 301 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	289.82	7.14	0.00895

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	289.82	12.43	0.01560

==========

COLUMN NO. 56 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.077 301 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	254.15	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	254.15	0.00	0.00000

==========

COLUMN NO. 57 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
AREA OF STEEL REQUIRED = 3524.5 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	308	STA	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	-347.45	76.71	-0.08028

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	-347.45	27.28	-0.02856

==========

COLUMN NO. 58 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4455.00   1204.46     270.4
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  485.93  -1583.37    267.08      0.00    0.00000
 ---------------------------------------------------------
```

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4260.51    834.62     195.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  337.39  -1583.37    267.08      0.00    0.00000
 ---------------------------------------------------------
```

==========

COLUMN NO. 59 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
AREA OF STEEL REQUIRED = 3916.5 SQ. MM

```
BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

  8 - 25 MM               1.122      317      STA     0.900
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 320.00 MM
```

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11945.00   9556.00   4406.92   1234.44     280.1
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  503.22  -1649.32   -939.24     19.28   -0.00746
 ---------------------------------------------------------
```

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11945.00   9556.00   4210.34    853.30     202.7
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  348.49  -1649.32   -939.24    154.64   -0.05987
 ---------------------------------------------------------
```

==========

COLUMN NO. 60 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	432.92	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	432.92	0.00	0.00000

==========

COLUMN NO. 61 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
AREA OF STEEL REQUIRED = 3818.5 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.122	318	STA	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11945.00	9556.00	4406.92	1234.44	280.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
503.22	-1649.32	-938.26	10.40	-0.00403

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11945.00	9556.00	4210.34	853.30	202.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
348.49	-1649.32	-938.26	149.84	-0.05807

==========

COLUMN NO. 62 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	435.08	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	435.08	0.00	0.00000

==========

COLUMN NO. 63 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	114.66	1.08	0.00630

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	114.66	15.97	0.09285

==========

COLUMN NO. 64 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
AREA OF STEEL REQUIRED = 2039.2 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	309	STA	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	-20.31	0.09	-0.00657

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	-20.31	2.13	-0.16110

==========

COLUMN NO. 65 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	125.61	3.19	0.01693

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	125.61	17.93	0.09518

==========

COLUMN NO. 66 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	122.77	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	122.77	27.62	0.34615

==========

COLUMN NO. 67 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	41.48	2.14	0.03443

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	41.48	7.73	0.12416

==========

COLUMN NO. 68 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	7.32	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	7.32	1.65	0.34615

==========

COLUMN NO. 69 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	33.92	0.41	0.00802

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
 6965.96   5572.77   2433.31    442.03     181.7
   M0        P-tens.   Des.Pn    Des.Mn     e/h
  191.29  -1013.36     33.92      5.60    0.11010
 ---------------------------------------------------------
```

==========

COLUMN NO. 70 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

```
BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

 12 - 16 MM               1.191      301      END     0.650
```

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
 6965.96   5572.77   2433.31    442.03     181.7
   M0        P-tens.   Des.Pn    Des.Mn     e/h
  191.29  -1013.36      7.32      0.00    0.00000
 ---------------------------------------------------------
```

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
 6965.96   5572.77   2433.31    442.03     181.7
   M0        P-tens.   Des.Pn    Des.Mn     e/h
  191.29  -1013.36      7.32      1.65    0.34615
 ---------------------------------------------------------
```

**********END OF COLUMN DESIGN RESULTS**********

880. DESIGN BEAM 81 TO 106

BEAM NO. 81 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 3400. MM FY - 420. FC - 35. MPA, SIZE - 500. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	5 - 12MM	0.	3400.	YES	YES

CRITICAL POS MOMENT= 40.09 KN-MET AT 0.MM, LOAD 315
REQD STEEL= 466.MM2, RHO=0.0040, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 92. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 15705.4 cm^4

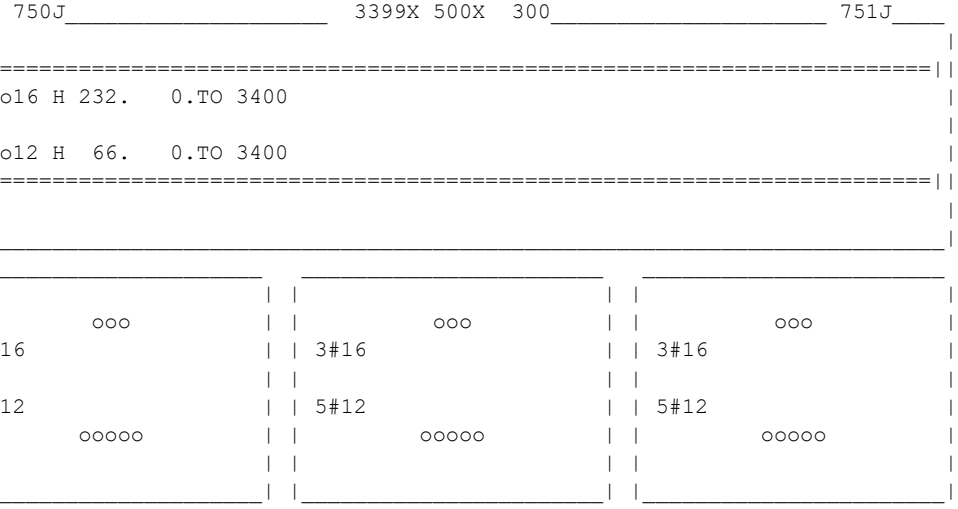
2	232.	3 - 16MM	0.	3400.	YES	YES

CRITICAL NEG MOMENT= 51.10 KN-MET AT 3400.MM, LOAD 307
REQD STEEL= 599.MM2, RHO=0.0051, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 182. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 16576.4 cm^4

B E A M N O. 81 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=    7.20 KNS  Vc=  114.17 KNS  Vs=    0.00 KNS
Tu=    0.07 KN-MET  Tc=    5.3 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=    7.35 KNS  Vc=  113.88 KNS  Vs=    0.00 KNS
Tu=    0.07 KN-MET  Tc=    5.3 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 82 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5300. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 12MM	0.	5300.	YES	YES
	CRITICAL POS MOMENT= 42.60 KN-MET AT 0.MM, LOAD 307					
	REQD STEEL= 283.MM2, RHO=0.0018, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 54977.0 cm^4					
2	534.	2 - 12MM	0.	2608.	YES	NO
	CRITICAL NEG MOMENT= 25.63 KN-MET AT 0.MM, LOAD 320					
	REQD STEEL= 170.MM2, RHO=0.0011, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 38198.0 cm^4					
3	534.	4 - 12MM	2471.	5300.	NO	YES

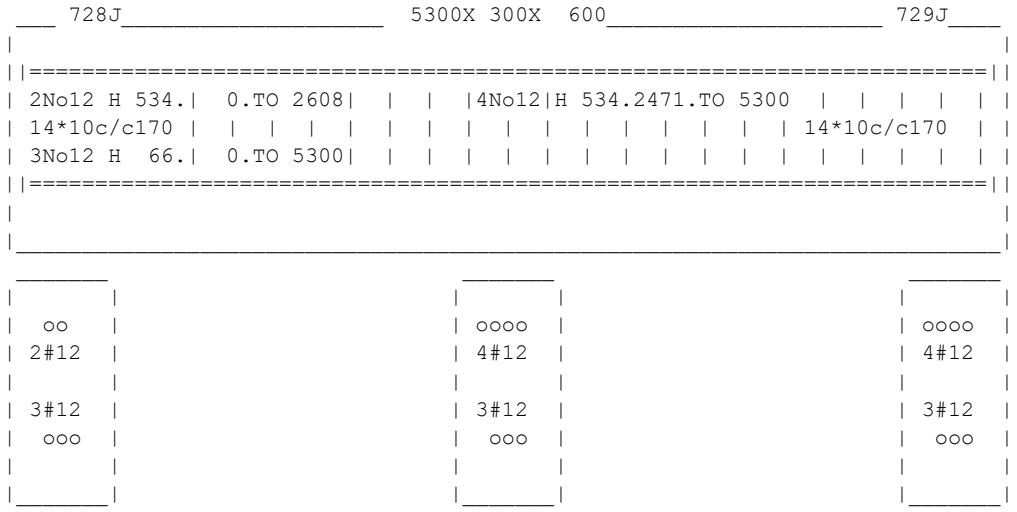
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|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     60.60 KN-MET  AT  5300.MM, LOAD  311|
|   REQD STEEL=   405.MM2, RHO=0.0025, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./   56. MMS        |
|   REQD. DEVELOPMENT LENGTH =   330. MMS                       |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 70807.2 cm^4

B E A M N O. 82 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 3.01 KNS Vc= 139.12 KNS Vs= 0.00 KNS
Tu= 14.14 KN-MET Tc= 5.9 KN-MET Ts= 18.9 KN-MET LOAD 302
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 170. MM C/C FOR 2120. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.86 SQ.CM.

AT END SUPPORT - Vu= 22.16 KNS Vc= 139.12 KNS Vs= 0.00 KNS
Tu= 14.14 KN-MET Tc= 5.9 KN-MET Ts= 18.9 KN-MET LOAD 302
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 170. MM C/C FOR 2120. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.86 SQ.CM.



BEAM NO. 83 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	5 - 12MM	626.	2000.	NO	YES

CRITICAL POS MOMENT= 79.54 KN-MET AT 2000.MM, LOAD 312
REQD STEEL= 534.MM2, RHO=0.0033, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 42. MMS
REQD. DEVELOPMENT LENGTH = 440. MMS

Cracked Moment of Inertia Iz at above location = 85862.6 cm^4

2	534.	5 - 12MM	0.	2000.	YES	YES

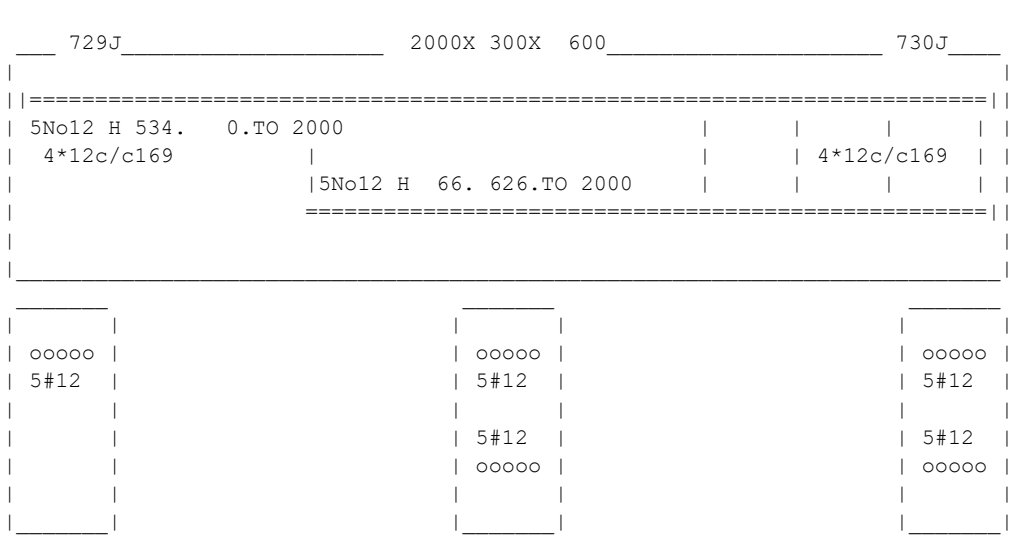
CRITICAL NEG MOMENT= 94.89 KN-MET AT 0.MM, LOAD 305
REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 42. MMS
REQD. DEVELOPMENT LENGTH = 440. MMS

Cracked Moment of Inertia Iz at above location = 85862.6 cm^4

B E A M N O. 83 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 87.62 KNS Vc= 155.74 KNS Vs= 0.00 KNS
Tu= 32.82 KN-MET Tc= 6.6 KN-MET Ts= 43.8 KN-MET LOAD 306
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 169. MM C/C FOR 470. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 8.95 SQ.CM.

AT END SUPPORT - Vu= 82.04 KNS Vc= 155.74 KNS Vs= 0.00 KNS
Tu= 32.82 KN-MET Tc= 6.6 KN-MET Ts= 43.8 KN-MET LOAD 306
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 169. MM C/C FOR 470. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 8.95 SQ.CM.



BEAM NO. 84 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	5 - 12MM	0.	5000.	YES	YES
	CRITICAL POS MOMENT= 87.38 KN-MET AT 5000.MM, LOAD 314					
	REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 42. MMS					
	REQD. DEVELOPMENT LENGTH = 440. MMS					
	Cracked Moment of Inertia Iz at above location = 85862.6 cm^4					
2	534.	5 - 12MM	0.	2483.	YES	NO
	CRITICAL NEG MOMENT= 93.21 KN-MET AT 0.MM, LOAD 314					
	REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 42. MMS					
	REQD. DEVELOPMENT LENGTH = 440. MMS					
	Cracked Moment of Inertia Iz at above location = 85862.6 cm^4					
3	534.	5 - 12MM	2308.	5000.	NO	YES

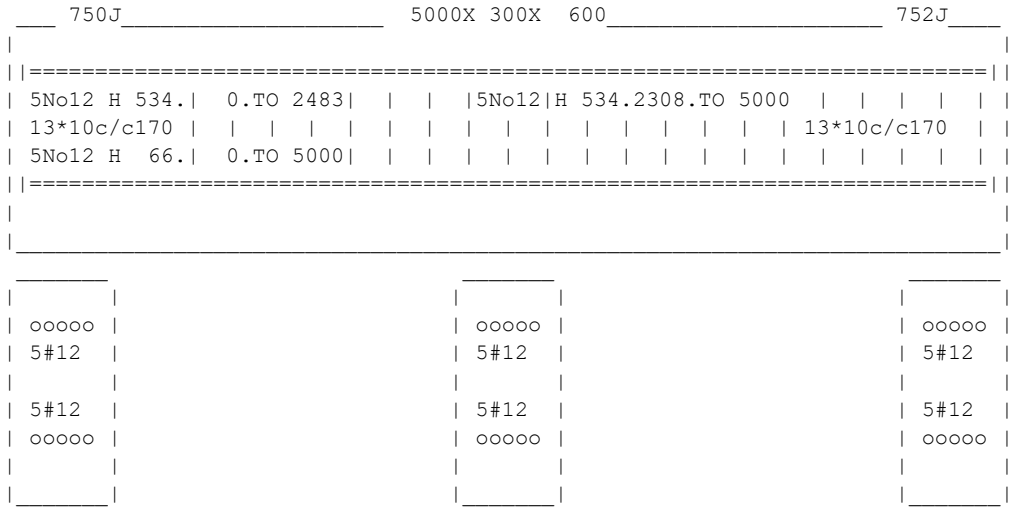
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|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     75.56 KN-MET  AT  5000.MM, LOAD  317|
|   REQD STEEL=   506.MM2, RHO=0.0032, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./   42. MMS         |
|   REQD. DEVELOPMENT LENGTH =   440. MMS                       |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 85862.6 cm^4

B E A M N O. 84 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 10.54 KNS Vc= 140.58 KNS Vs= 0.00 KNS
Tu= 10.40 KN-MET Tc= 6.0 KN-MET Ts= 13.9 KN-MET LOAD 312
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 170. MM C/C FOR 1970. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.84 SQ.CM.

AT END SUPPORT - Vu= 9.51 KNS Vc= 140.58 KNS Vs= 0.00 KNS
Tu= 10.40 KN-MET Tc= 6.0 KN-MET Ts= 13.9 KN-MET LOAD 312
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 170. MM C/C FOR 1970. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.84 SQ.CM.



BEAM NO. 85 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	5 - 12MM	0.	5000.	YES	YES

CRITICAL POS MOMENT=	96.78 KN-MET AT 5000.MM, LOAD 314
REQD STEEL=	562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING=	225./ 37./ 42. MMS
REQD. DEVELOPMENT LENGTH =	440. MMS

Cracked Moment of Inertia Iz at above location = 85862.6 cm^4

2	534.	5 - 12MM	0.	2483.	YES	NO

CRITICAL NEG MOMENT=	93.01 KN-MET AT 0.MM, LOAD 314
REQD STEEL=	562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING=	225./ 37./ 42. MMS
REQD. DEVELOPMENT LENGTH =	440. MMS

Cracked Moment of Inertia Iz at above location = 85862.6 cm^4

3	534.	5 - 12MM	2308.	5000.	NO	YES

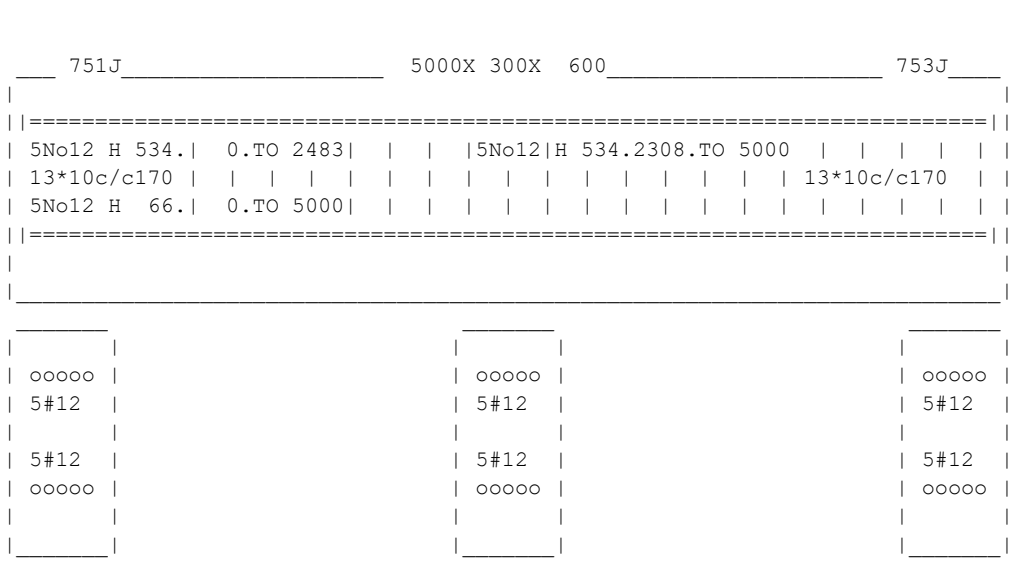
CRITICAL NEG MOMENT=	78.71 KN-MET AT 5000.MM, LOAD 317
REQD STEEL=	528.MM2, RHO=0.0033, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING=	225./ 37./ 42. MMS
REQD. DEVELOPMENT LENGTH =	440. MMS

Cracked Moment of Inertia Iz at above location = 85862.6 cm^4

B E A M N O. 85 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 8.49 KNS Vc= 149.43 KNS Vs= 0.00 KNS
Tu= 10.20 KN-MET Tc= 6.3 KN-MET Ts= 13.6 KN-MET LOAD 320
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 170. MM C/C FOR 1970. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.78 SQ.CM.

AT END SUPPORT - Vu= 6.55 KNS Vc= 149.43 KNS Vs= 0.00 KNS
Tu= 10.20 KN-MET Tc= 6.3 KN-MET Ts= 13.6 KN-MET LOAD 320
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 170. MM C/C FOR 1970. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.78 SQ.CM.



BEAM NO. 86 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5850. MM FY - 420. FC - 35. MPA, SIZE - 500. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	4 - 12MM	939.	5850.	NO	YES

CRITICAL POS MOMENT= 33.86 KN-MET AT 5850.MM, LOAD 314
REQD STEEL= 411.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 123. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 13015.0 cm^4

2	234.	7 - 12MM	0.	2745.	YES	NO

CRITICAL NEG MOMENT= 58.80 KN-MET AT 0.MM, LOAD 314
REQD STEEL= 694.MM2, RHO=0.0059, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 61. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 20693.5 cm^4

3	234.	3 - 12MM	4811.	5850.	NO YES

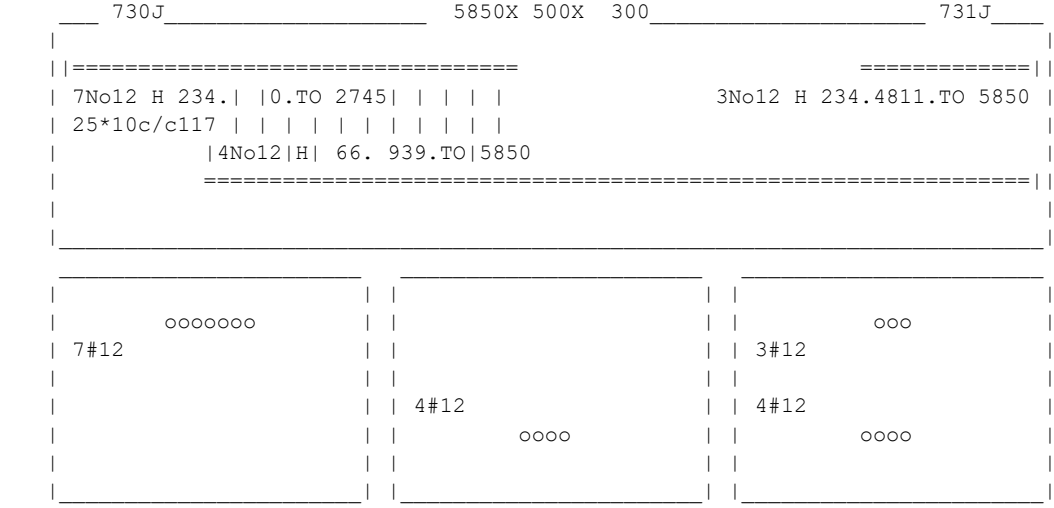
CRITICAL NEG MOMENT= 6.98 KN-MET AT 5850.MM, LOAD 317
REQD STEEL= 105.MM2, RHO=0.0009, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 184. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 10162.3 cm^4

B E A M N O. 86 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 27.89 KNS Vc= 69.32 KNS Vs= 0.00 KNS
Tu= 1.79 KN-MET Tc= 2.9 KN-MET Ts= 0.0 KN-MET LOAD 302
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 117. MM C/C FOR 2695. MM

AT END SUPPORT - Vu= 6.70 KNS Vc= 81.57 KNS Vs= 0.00 KNS
Tu= 0.61 KN-MET Tc= 3.6 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 87 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 500. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	4 - 12MM	1946.	6000.	NO	YES

CRITICAL POS MOMENT=	38.40 KN-MET AT 6000.MM, LOAD 305
REQD STEEL=	446.MM2, RHO=0.0038, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING=	225./ 37./ 123. MMS
REQD. DEVELOPMENT LENGTH =	308. MMS

Cracked Moment of Inertia Iz at above location = 13015.0 cm^4

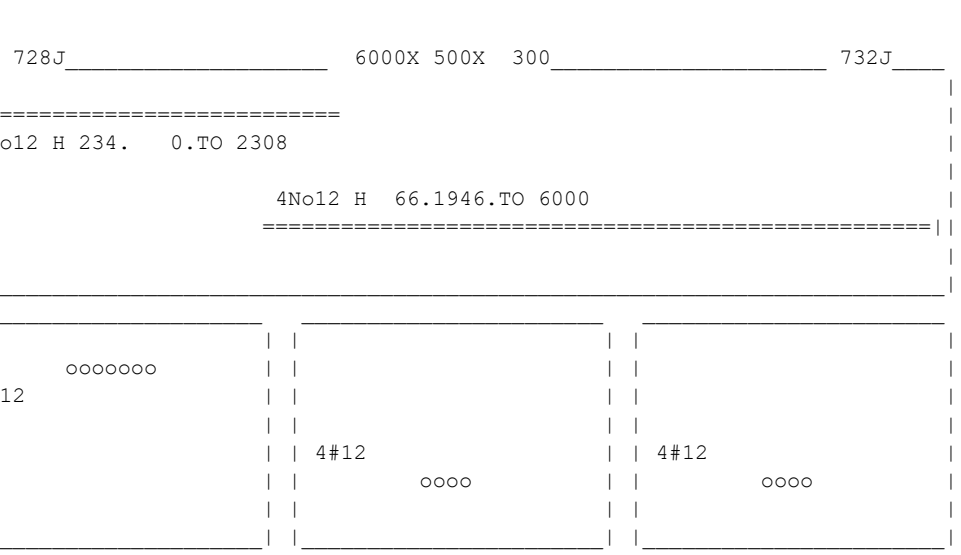
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
2	234.	7 - 12MM	0.	2308.	YES	NO

CRITICAL NEG MOMENT=	60.11 KN-MET AT 0.MM, LOAD 305
REQD STEEL=	710.MM2, RHO=0.0061, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING=	225./ 37./ 61. MMS
REQD. DEVELOPMENT LENGTH =	308. MMS

Cracked Moment of Inertia Iz at above location = 20693.5 cm^4

B E A M N O. 87 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 23.80 KNS Vc= 110.46 KNS Vs= 0.00 KNS
Tu= 0.51 KN-MET Tc= 5.4 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 3.61 KNS Vc= 109.60 KNS Vs= 0.00 KNS
Tu= 0.51 KN-MET Tc= 5.4 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 88 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 3400. MM FY - 420. FC - 35. MPA, SIZE - 500. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	4 - 12MM	0.	3400.	YES	YES

CRITICAL POS MOMENT= 37.26 KN-MET AT 0.MM, LOAD 319
REQD STEEL= 433.MM2, RHO=0.0037, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 123. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 13015.0 cm^4

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
2	234.	6 - 12MM	0.	3400.	YES	YES

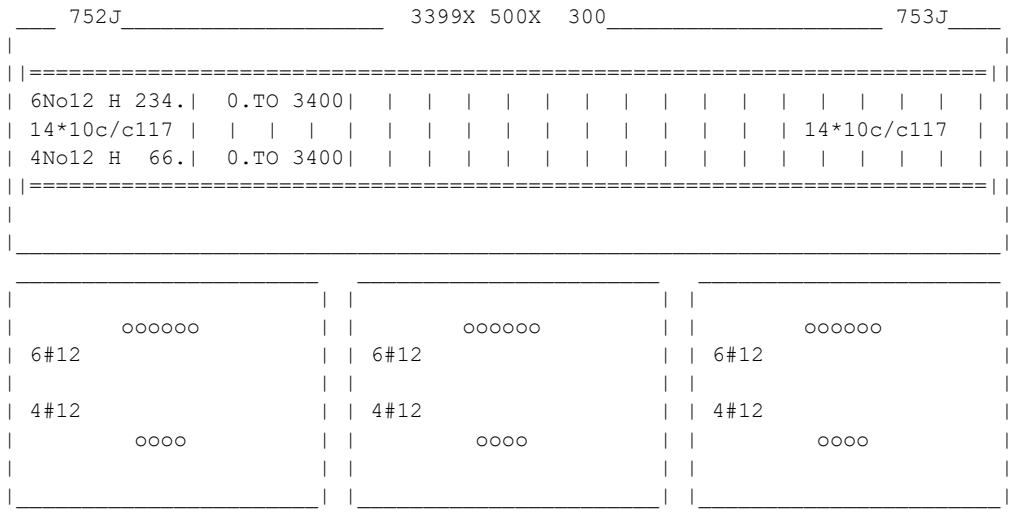
CRITICAL NEG MOMENT= 55.83 KN-MET AT 3400.MM, LOAD 311
REQD STEEL= 657.MM2, RHO=0.0056, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 74. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 18258.8 cm^4

B E A M N O. 88 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.20 KNS Vc= 111.29 KNS Vs= 0.00 KNS
Tu= 6.34 KN-MET Tc= 5.3 KN-MET Ts= 8.5 KN-MET LOAD 308
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 117. MM C/C FOR 1470. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.85 SQ.CM.

AT END SUPPORT - Vu= 1.26 KNS Vc= 110.08 KNS Vs= 0.00 KNS
Tu= 6.34 KN-MET Tc= 5.3 KN-MET Ts= 8.5 KN-MET LOAD 308
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 117. MM C/C FOR 1470. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.85 SQ.CM.



BEAM NO. 89 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5302. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 12MM	0.	5302.	YES	YES

CRITICAL POS MOMENT= 49.21 KN-MET AT 5302.MM, LOAD 320
REQD STEEL= 328.MM2, RHO=0.0020, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 84. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 54977.0 cm^4

2	532.	2 - 16MM	0.	2743.	YES	NO

CRITICAL NEG MOMENT= 57.39 KN-MET AT 0.MM, LOAD 312
REQD STEEL= 384.MM2, RHO=0.0024, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 164. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 63391.8 cm^4

3	534.	3 - 12MM	2472.	5302.	NO	YES

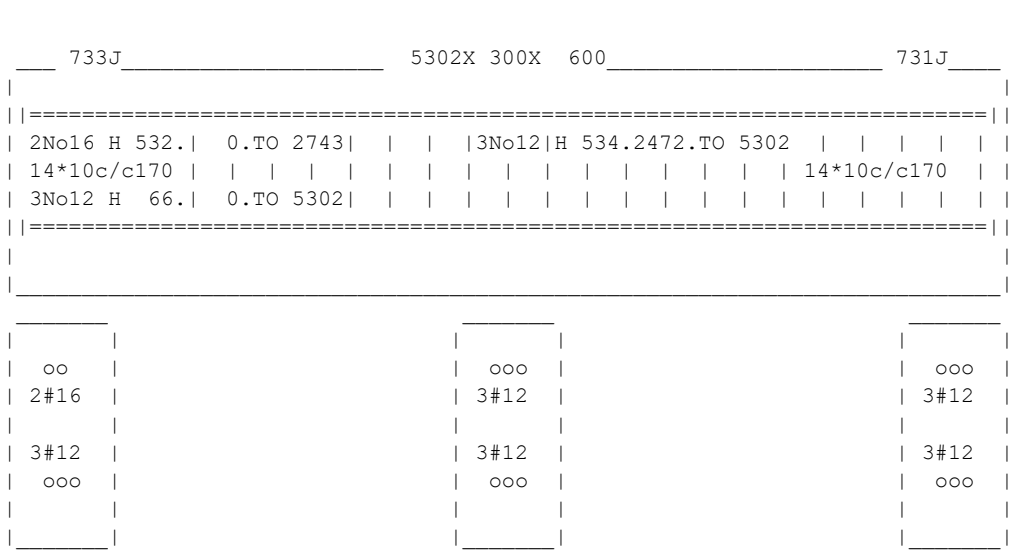
CRITICAL NEG MOMENT= 41.87 KN-MET AT 5302.MM, LOAD 307
REQD STEEL= 278.MM2, RHO=0.0017, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 84. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 54977.0 cm^4

B E A M N O. 89 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 15.40 KNS Vc= 150.90 KNS Vs= 0.00 KNS
Tu= 15.21 KN-MET Tc= 6.4 KN-MET Ts= 20.3 KN-MET LOAD 306
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 170. MM C/C FOR 2121. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 4.15 SQ.CM.

AT END SUPPORT - Vu= 9.79 KNS Vc= 150.90 KNS Vs= 0.00 KNS
Tu= 15.21 KN-MET Tc= 6.4 KN-MET Ts= 20.3 KN-MET LOAD 306
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 170. MM C/C FOR 2121. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 4.15 SQ.CM.



BEAM NO. 90 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	4 - 12MM	0.	1275.	YES	NO

CRITICAL POS MOMENT= 60.55 KN-MET AT 0.MM, LOAD 307
REQD STEEL= 404.MM2, RHO=0.0025, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 56. MMS
REQD. DEVELOPMENT LENGTH = 330. MMS

Cracked Moment of Inertia Iz at above location = 70807.2 cm^4

2	534.	4 - 12MM	0.	2000.	YES	YES

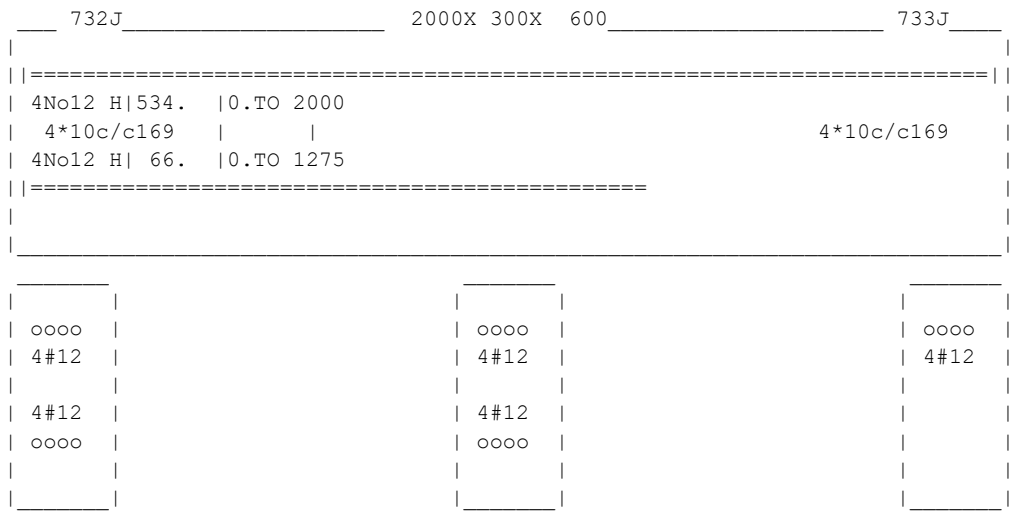
CRITICAL NEG MOMENT= 64.78 KN-MET AT 2000.MM, LOAD 307
REQD STEEL= 433.MM2, RHO=0.0027, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 56. MMS
REQD. DEVELOPMENT LENGTH = 330. MMS

Cracked Moment of Inertia Iz at above location = 70807.2 cm^4

B E A M N O. 90 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 8.78 KNS Vc= 278.29 KNS Vs= 0.00 KNS
Tu= 22.76 KN-MET Tc= 6.8 KN-MET Ts= 30.4 KN-MET LOAD 305
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 169. MM C/C FOR 470. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 6.21 SQ.CM.

AT END SUPPORT - Vu= 14.36 KNS Vc= 158.15 KNS Vs= 0.00 KNS
Tu= 22.76 KN-MET Tc= 6.8 KN-MET Ts= 30.4 KN-MET LOAD 305
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 169. MM C/C FOR 470. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 6.21 SQ.CM.



BEAM NO. 91 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 6030. MM FY - 420. FC - 35. MPA, SIZE - 500. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 12MM	0.	6030.	YES	YES

CRITICAL POS MOMENT= 15.20 KN-MET AT 5025.MM, LOAD 308
REQD STEEL= 231.MM2, RHO=0.0020, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 184. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 10162.3 cm^4

2	234.	4 - 12MM	0.	2318.	YES	NO

CRITICAL NEG MOMENT= 37.40 KN-MET AT 0.MM, LOAD 308
REQD STEEL= 434.MM2, RHO=0.0037, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 123. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 13015.0 cm^4

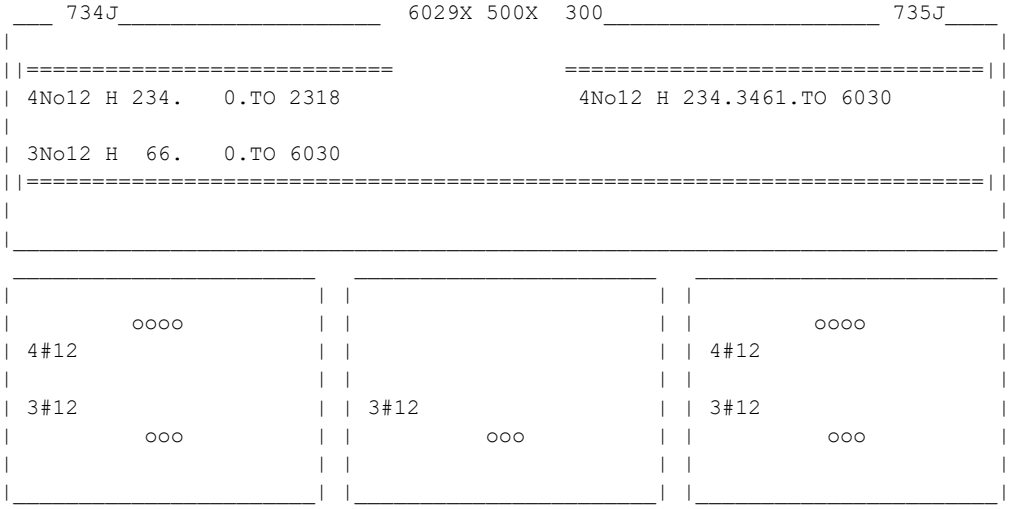
3	234.	4 - 12MM	3461.	6030.	NO	YES

CRITICAL NEG MOMENT= 30.46 KN-MET AT 6030.MM, LOAD 311
REQD STEEL= 411.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 123. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 13015.0 cm^4

B E A M N O. 91 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 16.19 KNS Vc= 84.97 KNS Vs= 0.00 KNS
Tu= 1.28 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 11.37 KNS Vc= 84.97 KNS Vs= 0.00 KNS
Tu= 1.28 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 92 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5830. MM FY - 420. FC - 35. MPA, SIZE - 500. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 12MM	0.	5830.	YES	YES

CRITICAL POS MOMENT= 13.13 KN-MET AT 4858.MM, LOAD 316
REQD STEEL= 199.MM2, RHO=0.0017, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 184. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 10162.3 cm^4

2	234.	4 - 12MM	0.	2251.	YES	NO

CRITICAL NEG MOMENT= 30.24 KN-MET AT 0.MM, LOAD 308
REQD STEEL= 411.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 123. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 13015.0 cm^4

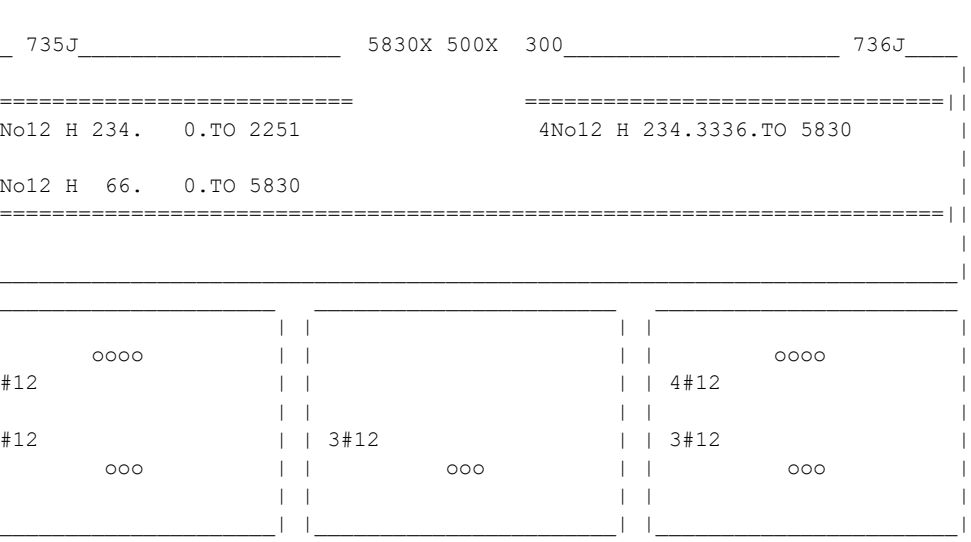
3	234.	4 - 12MM	3336.	5830.	NO	YES

CRITICAL NEG MOMENT= 28.40 KN-MET AT 5830.MM, LOAD 311
REQD STEEL= 411.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 123. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 13015.0 cm^4

B E A M N O. 92 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 13.05 KNS Vc= 112.64 KNS Vs= 0.00 KNS
Tu= 0.50 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 13.52 KNS Vc= 112.64 KNS Vs= 0.00 KNS
Tu= 0.50 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 93 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 7500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	68.	4 - 16MM	0.	7500.	YES	YES

CRITICAL POS MOMENT= 134.59 KN-MET AT 0.MM, LOAD 309
REQD STEEL= 690.MM2, RHO=0.0043, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 55. MMS
REQD. DEVELOPMENT LENGTH = 601. MMS

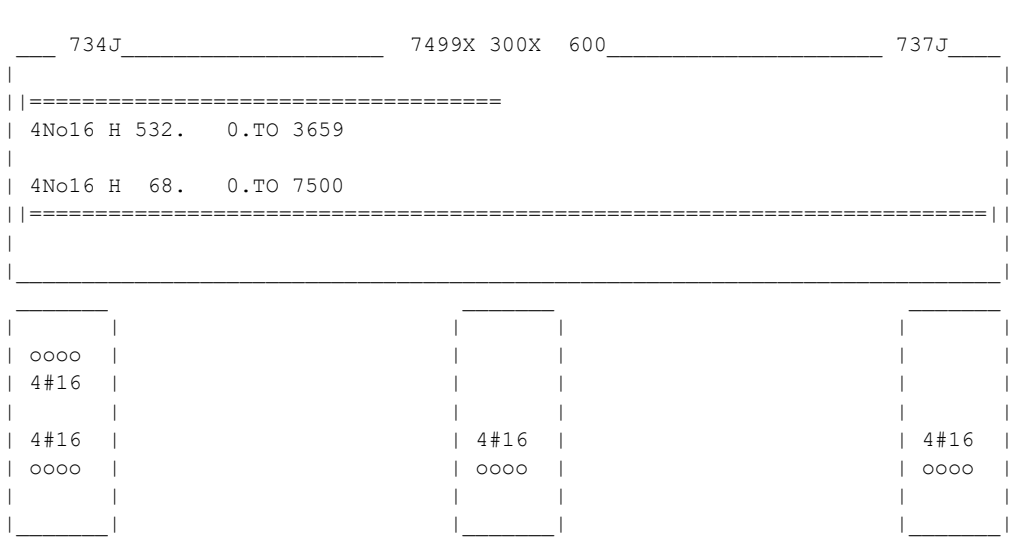
Cracked Moment of Inertia Iz at above location = 114685.2 cm^4

2	532.	4 - 16MM	0.	3659.	YES	NO

CRITICAL NEG MOMENT= 133.37 KN-MET AT 0.MM, LOAD 322
REQD STEEL= 684.MM2, RHO=0.0043, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 55. MMS
REQD. DEVELOPMENT LENGTH = 601. MMS

Cracked Moment of Inertia Iz at above location = 114685.2 cm^4

*** A SUITABLE BAR ARRANGEMENT COULD NOT BE DETERMINED.
REQD. STEEL = 975. MM2, MAX. STEEL PERMISSIBLE = 3924. MM2
MAX NEG MOMENT = 185.30 KN-MET, LOADING 309



BEAM NO. 94 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 7500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	5 - 12MM	0.	7500.	YES	YES
	CRITICAL POS MOMENT= 101.80 KN-MET AT 0.MM, LOAD 321					
	REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 42. MMS					
	REQD. DEVELOPMENT LENGTH = 440. MMS					
	Cracked Moment of Inertia Iz at above location = 85862.6 cm^4					
2	532.	4 - 16MM	0.	3659.	YES	NO
	CRITICAL NEG MOMENT= 122.00 KN-MET AT 0.MM, LOAD 310					
	REQD STEEL= 624.MM2, RHO=0.0039, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 55. MMS					
	REQD. DEVELOPMENT LENGTH = 601. MMS					
	Cracked Moment of Inertia Iz at above location = 114685.2 cm^4					
3	532.	4 - 16MM	3529.	7500.	NO	YES

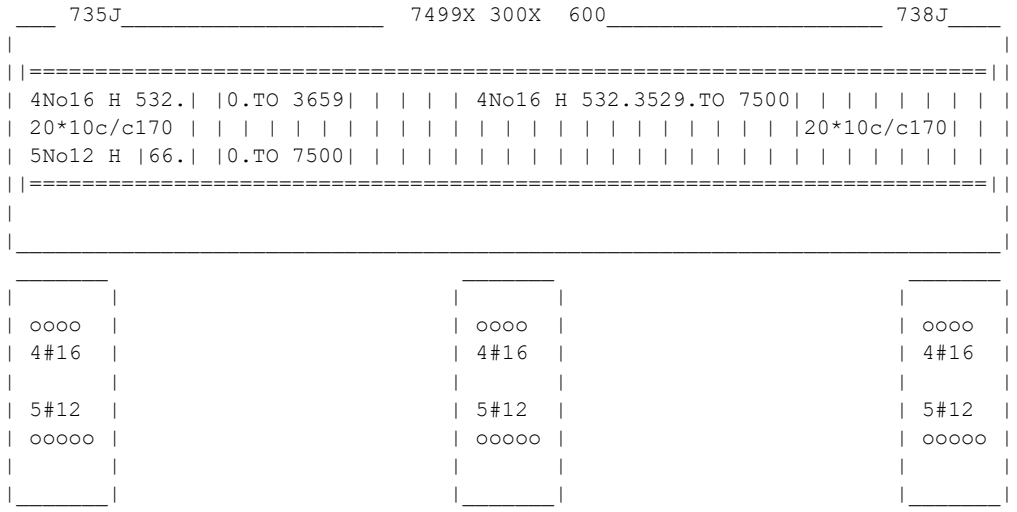
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|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=    151.51 KN-MET  AT  7500.MM, LOAD  313|
|   REQD STEEL=   780.MM2, RHO=0.0049, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   41./   55. MMS         |
|   REQD. DEVELOPMENT LENGTH =   601. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =   114685.2 cm^4
```

B E A M N O. 94 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.36 KNS Vc= 106.56 KNS Vs= 0.00 KNS
Tu= 5.93 KN-MET Tc= 4.4 KN-MET Ts= 7.9 KN-MET LOAD 307
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 170. MM C/C FOR 3220. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.62 SQ.CM.

AT END SUPPORT - Vu= 20.42 KNS Vc= 106.56 KNS Vs= 0.00 KNS
Tu= 5.93 KN-MET Tc= 4.4 KN-MET Ts= 7.9 KN-MET LOAD 307
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 170. MM C/C FOR 3220. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.62 SQ.CM.



BEAM NO. 95 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 7503. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	5 - 12MM	0.	7503.	YES	YES

CRITICAL POS MOMENT= 92.77 KN-MET AT 0.MM, LOAD 321
REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 42. MMS
REQD. DEVELOPMENT LENGTH = 440. MMS

Cracked Moment of Inertia Iz at above location = 85862.6 cm^4

2	532.	4 - 16MM	0.	3660.	YES	NO

CRITICAL NEG MOMENT= 123.67 KN-MET AT 0.MM, LOAD 310
REQD STEEL= 633.MM2, RHO=0.0040, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 55. MMS
REQD. DEVELOPMENT LENGTH = 601. MMS

Cracked Moment of Inertia Iz at above location = 114685.2 cm^4

3	532.	4 - 16MM	3530.	7503.	NO	YES

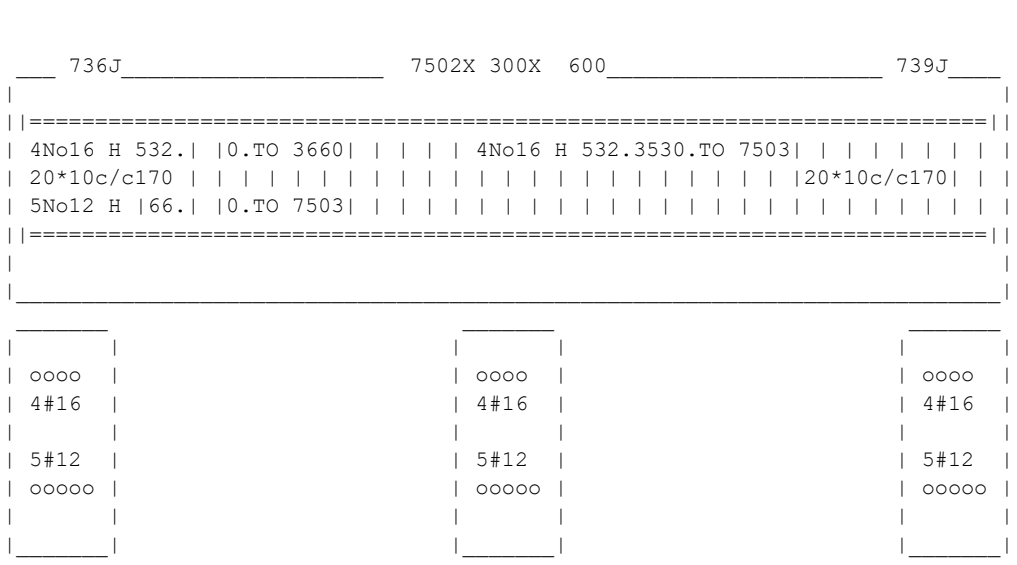
CRITICAL NEG MOMENT= 138.54 KN-MET AT 7503.MM, LOAD 313
REQD STEEL= 711.MM2, RHO=0.0045, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 55. MMS
REQD. DEVELOPMENT LENGTH = 601. MMS

Cracked Moment of Inertia Iz at above location = 114685.2 cm^4

B E A M N O. 95 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 15.84 KNS Vc= 118.30 KNS Vs= 0.00 KNS
Tu= 8.73 KN-MET Tc= 5.0 KN-MET Ts= 11.6 KN-MET LOAD 312
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 170. MM C/C FOR 3221. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.38 SQ.CM.

AT END SUPPORT - Vu= 16.95 KNS Vc= 118.30 KNS Vs= 0.00 KNS
Tu= 8.73 KN-MET Tc= 5.0 KN-MET Ts= 11.6 KN-MET LOAD 312
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 170. MM C/C FOR 3221. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.38 SQ.CM.



BEAM NO. 96 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 6030. MM FY - 420. FC - 35. MPA, SIZE - 500. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	5 - 12MM	0.	6030.	YES	YES

CRITICAL POS MOMENT= 43.01 KN-MET AT 6030.MM, LOAD 316
REQD STEEL= 501.MM2, RHO=0.0043, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 92. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 15705.4 cm^4

2	234.	7 - 12MM	0.	2820.	YES	NO

CRITICAL NEG MOMENT= 65.80 KN-MET AT 0.MM, LOAD 308
REQD STEEL= 781.MM2, RHO=0.0067, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 61. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 20693.5 cm^4

3	234.	7 - 12MM	2958.	6030.	NO YES

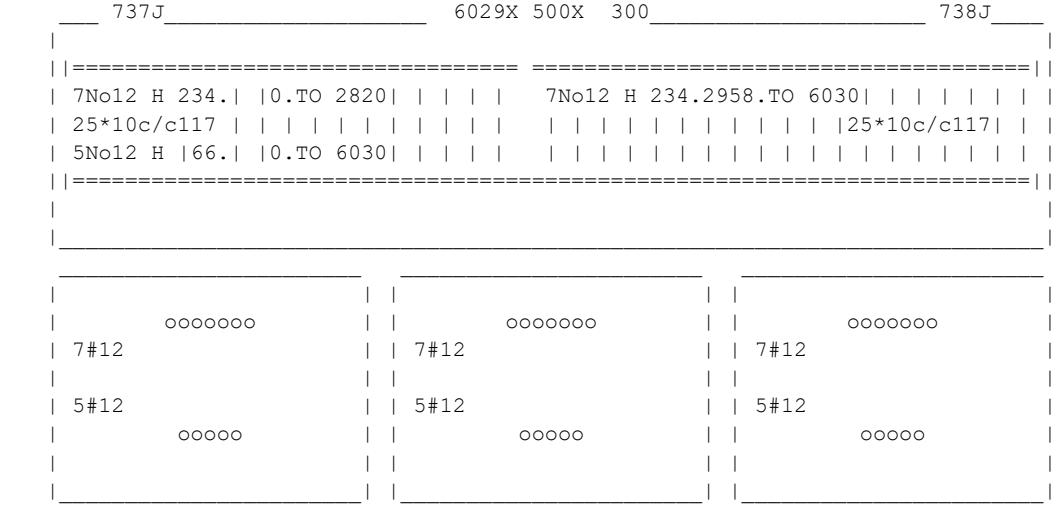
CRITICAL NEG MOMENT= 64.07 KN-MET AT 6030.MM, LOAD 311
REQD STEEL= 759.MM2, RHO=0.0065, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 61. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 20693.5 cm^4

B E A M N O. 96 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 4.17 KNS Vc= 47.85 KNS Vs= 0.00 KNS
Tu= 2.19 KN-MET Tc= NaN KN-MET Ts= 2.9 KN-MET LOAD 307
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 117. MM C/C FOR 2785. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 0.64 SQ.CM.

AT END SUPPORT - Vu= 27.79 KNS Vc= 47.85 KNS Vs= 0.00 KNS
Tu= 2.19 KN-MET Tc= NaN KN-MET Ts= 2.9 KN-MET LOAD 307
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 117. MM C/C FOR 2785. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 0.64 SQ.CM.



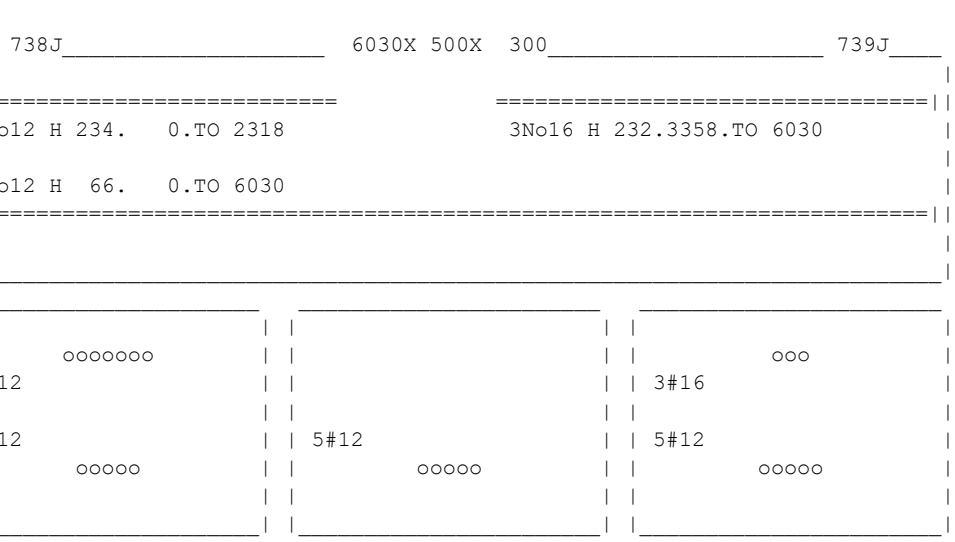
BEAM NO. 97 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 6030. MM FY - 420. FC - 35. MPA, SIZE - 500. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	5 - 12MM	0.	6030.	YES	YES
	CRITICAL POS MOMENT= 42.79 KN-MET AT 6030.MM, LOAD 316					
	REQD STEEL= 499.MM2, RHO=0.0043, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 92. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 15705.4 cm^4					
2	234.	7 - 12MM	0.	2318.	YES	NO
	CRITICAL NEG MOMENT= 57.76 KN-MET AT 0.MM, LOAD 308					
	REQD STEEL= 681.MM2, RHO=0.0058, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 61. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 20693.5 cm^4					
3	232.	3 - 16MM	3358.	6030.	NO	YES
	CRITICAL NEG MOMENT= 48.60 KN-MET AT 6030.MM, LOAD 311					
	REQD STEEL= 574.MM2, RHO=0.0050, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 182. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 16270.0 cm^4					

B E A M N O. 97 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 15.16 KNS Vc= 111.17 KNS Vs= 0.00 KNS
Tu= 0.80 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 12.40 KNS Vc= 112.41 KNS Vs= 0.00 KNS
Tu= 0.80 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 98 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 12MM	0.	1500.	YES	YES

CRITICAL POS MOMENT= 16.25 KN-MET AT 0.MM, LOAD 307
REQD STEEL= 246.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 84. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 9423.2 cm^4

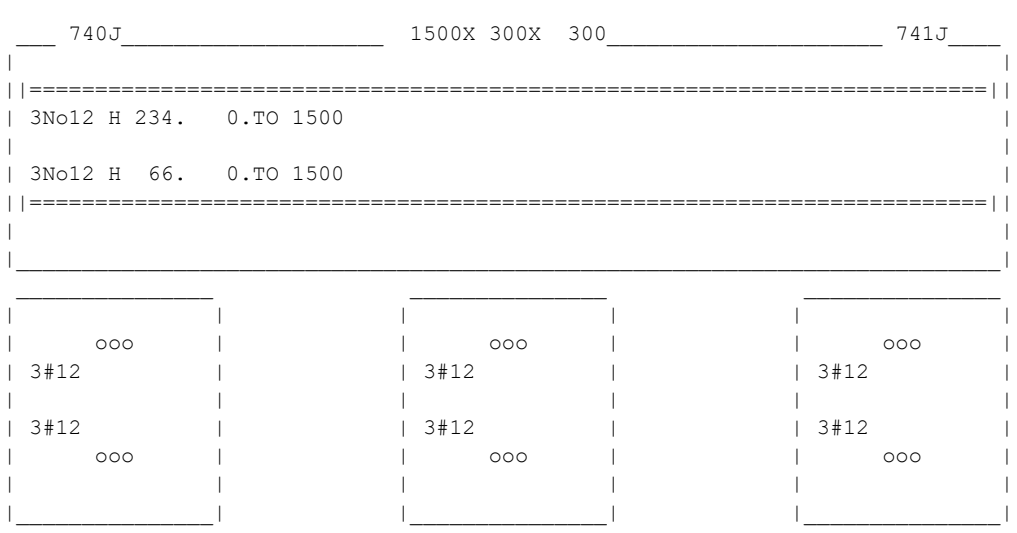
2	234.	3 - 12MM	0.	1500.	YES	YES

CRITICAL NEG MOMENT= 19.48 KN-MET AT 1500.MM, LOAD 307
REQD STEEL= 246.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 84. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 9423.2 cm^4

B E A M N O. 98 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 2.50 KNS Vc= 125.39 KNS Vs= 0.00 KNS
Tu= 0.12 KN-MET Tc= 2.7 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 5.58 KNS Vc= 72.37 KNS Vs= 0.00 KNS
Tu= 0.12 KN-MET Tc= 2.7 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 99 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 12MM	0.	838.	YES	NO
	CRITICAL POS MOMENT= 15.33 KN-MET AT 0.MM, LOAD 319					
	REQD STEEL= 235.MM2, RHO=0.0033, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 9423.2 cm^4					
2	66.	3 - 12MM	589.	1500.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL POS MOMENT=     16.24 KN-MET  AT  1500.MM, LOAD  316|
|   REQD STEEL=   246.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./   84. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|-------------------------------------------------------------|
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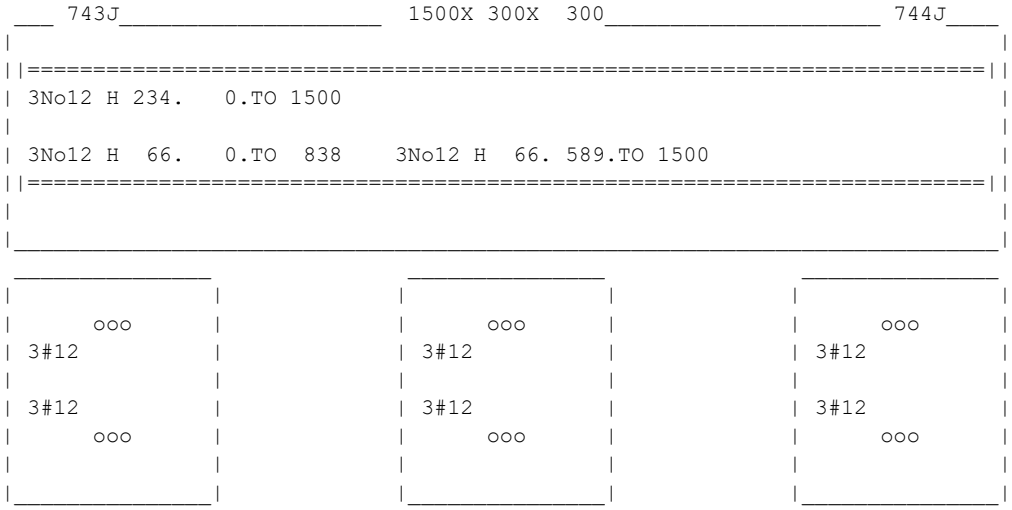
Cracked Moment of Inertia Iz at above location = 9423.2 cm^4

```
  3        234.     3 - 12MM         0.         1500.        YES  YES
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     19.82 KN-MET  AT     0.MM, LOAD  308|
|   REQD STEEL=   246.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./   84. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 9423.2 cm^4

B E A M N O. 99 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=    2.99 KNS  Vc=  125.02 KNS  Vs=    0.00 KNS
Tu=    0.18 KN-MET  Tc=    2.6 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=    0.10 KNS  Vc=  125.02 KNS  Vs=    0.00 KNS
Tu=    0.18 KN-MET  Tc=    2.6 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 100 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

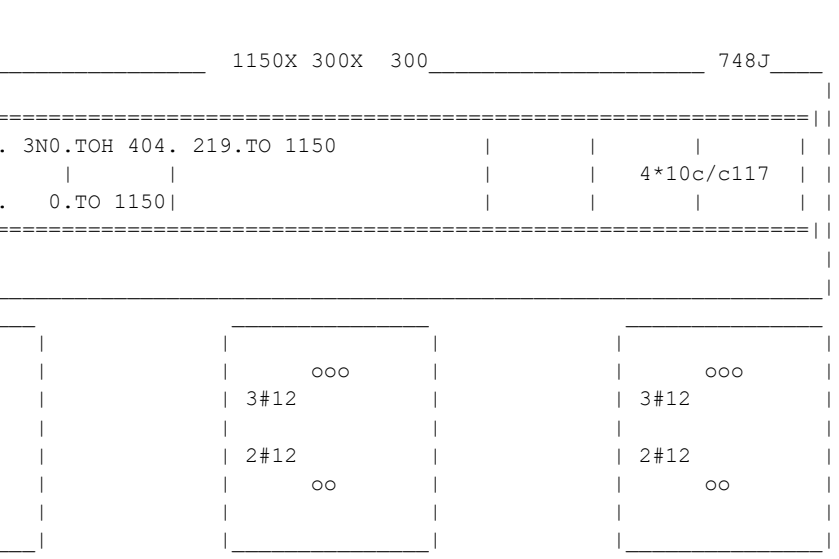
LEN - 1150. MM FY - 420. FC - 35. MPA, SIZE - 300. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 12MM	0.	1150.	YES	YES
	CRITICAL POS MOMENT= 13.29 KN-MET AT 0.MM, LOAD 309					
	REQD STEEL= 203.MM2, RHO=0.0029, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 6680.0 cm^4					
2	234.	2 - 12MM	0.	404.	YES	NO
	CRITICAL NEG MOMENT= 1.60 KN-MET AT 0.MM, LOAD 322					
	REQD STEEL= 24.MM2, RHO=0.0003, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 6680.0 cm^4					
3	234.	3 - 12MM	219.	1150.	NO	YES
	CRITICAL NEG MOMENT= 17.27 KN-MET AT 1150.MM, LOAD 313					
	REQD STEEL= 246.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 9423.2 cm^4					

B E A M N O. 100 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 25.70 KNS Vc= 66.57 KNS Vs= 0.00 KNS
Tu= 0.54 KN-MET Tc= 2.4 KN-MET Ts= 0.0 KN-MET LOAD 309
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 117. MM C/C FOR 345. MM

AT END SUPPORT - Vu= 27.46 KNS Vc= 66.57 KNS Vs= 0.00 KNS
Tu= 0.54 KN-MET Tc= 2.4 KN-MET Ts= 0.0 KN-MET LOAD 309
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 117. MM C/C FOR 345. MM



BEAM NO. 101 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1150. MM FY - 420. FC - 35. MPA, SIZE - 300. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 12MM	0.	1150.	YES	YES

CRITICAL POS MOMENT= 8.27 KN-MET AT 0.MM, LOAD 307
REQD STEEL= 126.MM2, RHO=0.0018, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 168. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 6680.0 cm^4

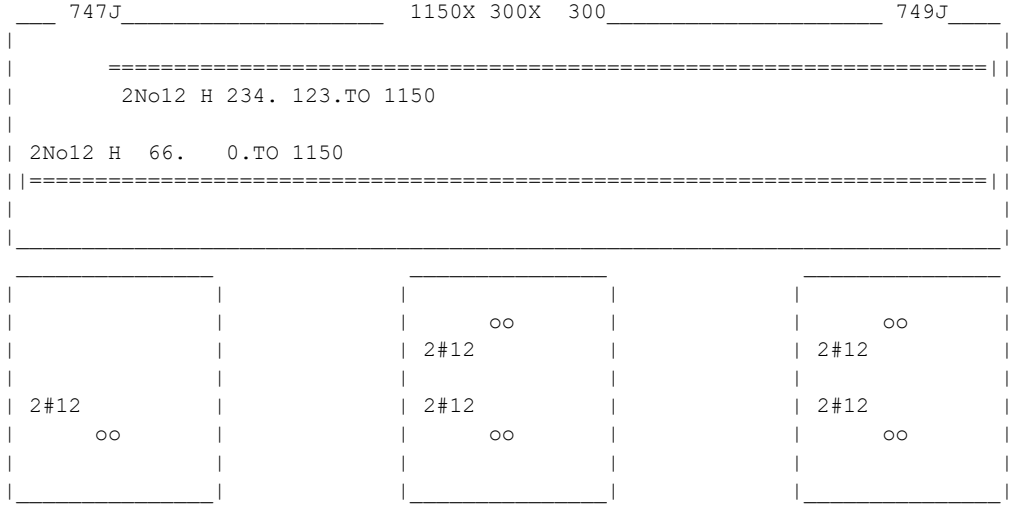
2	234.	2 - 12MM	123.	1150.	NO	YES

CRITICAL NEG MOMENT= 9.82 KN-MET AT 1150.MM, LOAD 313
REQD STEEL= 149.MM2, RHO=0.0021, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 168. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 6680.0 cm^4

B E A M N O. 101 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 8.18 KNS Vc= 66.78 KNS Vs= 0.00 KNS
Tu= 0.16 KN-MET Tc= 2.5 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 10.23 KNS Vc= 70.15 KNS Vs= 0.00 KNS
Tu= 0.16 KN-MET Tc= 2.5 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 102 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	5 - 12MM	0.	1112.	YES	NO

CRITICAL POS MOMENT= 62.21 KN-MET AT 0.MM, LOAD 317
REQD STEEL= 511.MM2, RHO=0.0051, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 42. MMS
REQD. DEVELOPMENT LENGTH = 440. MMS

Cracked Moment of Inertia Iz at above location = 31152.1 cm^4

2	66.	5 - 12MM	818.	2000.	NO	YES

CRITICAL POS MOMENT= 59.33 KN-MET AT 2000.MM, LOAD 322
REQD STEEL= 487.MM2, RHO=0.0049, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 42. MMS
REQD. DEVELOPMENT LENGTH = 440. MMS

Cracked Moment of Inertia Iz at above location = 31152.1 cm^4

3	332.	3 - 16MM	0.	2000.	YES	YES

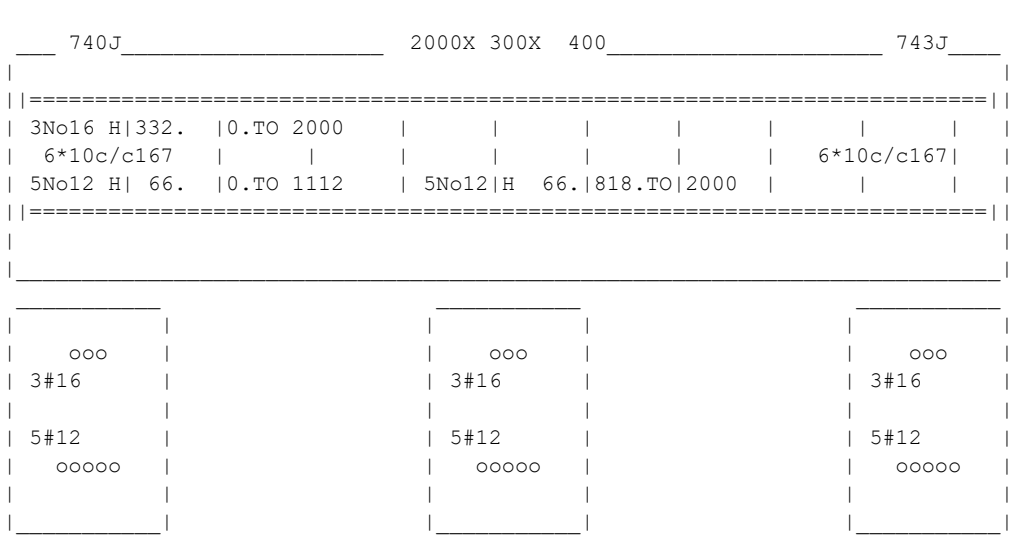
CRITICAL NEG MOMENT= 70.37 KN-MET AT 2000.MM, LOAD 309
REQD STEEL= 585.MM2, RHO=0.0059, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 82. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 32424.0 cm^4

B E A M N O. 102 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 63.91 KNS Vc= 98.16 KNS Vs= 0.00 KNS
Tu= 0.13 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 309
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 167. MM C/C FOR 670. MM

AT END SUPPORT - Vu= 68.45 KNS Vc= 97.77 KNS Vs= 0.00 KNS
Tu= 0.13 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 309
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 167. MM C/C FOR 670. MM



BEAM NO. 103 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	4 - 12MM	0.	1135.	YES	NO
	CRITICAL POS MOMENT= 54.22 KN-MET AT 0.MM, LOAD 321					
	REQD STEEL= 443.MM2, RHO=0.0044, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 56. MMS					
	REQD. DEVELOPMENT LENGTH = 330. MMS					
	Cracked Moment of Inertia Iz at above location = 25884.9 cm^4					
2	66.	5 - 12MM	819.	2000.	NO	YES
	CRITICAL POS MOMENT= 59.18 KN-MET AT 2000.MM, LOAD 318					
	REQD STEEL= 485.MM2, RHO=0.0048, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 42. MMS					
	REQD. DEVELOPMENT LENGTH = 440. MMS					
	Cracked Moment of Inertia Iz at above location = 31152.1 cm^4					
3	334.	5 - 12MM	0.	2000.	YES	YES

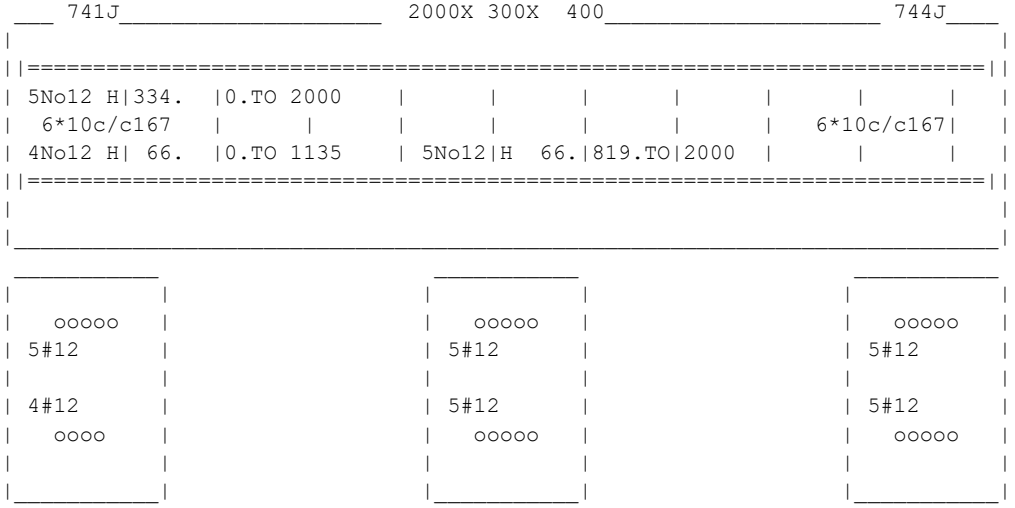
```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     66.55 KN-MET  AT     0.MM, LOAD  310|
|   REQD STEEL=   548.MM2, RHO=0.0055, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./   42. MMS         |
|   REQD. DEVELOPMENT LENGTH =   440. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    31152.1 cm^4
```

B E A M N O. 103 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   53.94 KNS  Vc=   98.25 KNS  Vs=    0.00 KNS
Tu=    1.39 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   309
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 167. MM  C/C FOR  670. MM

AT END   SUPPORT - Vu=   58.49 KNS  Vc=   97.78 KNS  Vs=    0.00 KNS
Tu=    1.39 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   309
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 167. MM  C/C FOR  670. MM
```



BEAM NO. 104 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

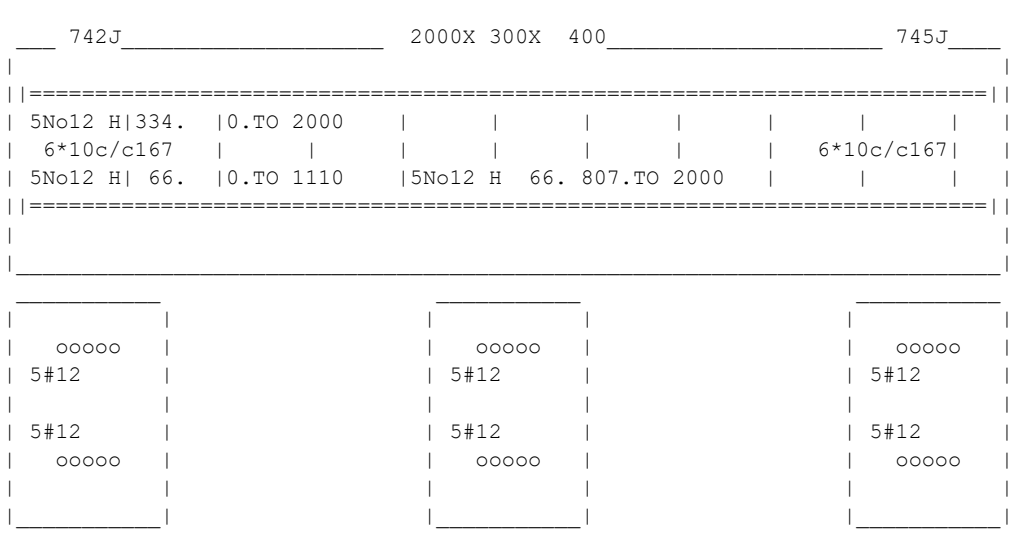
LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	5 - 12MM	0.	1110.	YES	NO
	CRITICAL POS MOMENT= 61.96 KN-MET AT 0.MM, LOAD 317					
	REQD STEEL= 509.MM2, RHO=0.0051, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 42. MMS					
	REQD. DEVELOPMENT LENGTH = 440. MMS					
	Cracked Moment of Inertia Iz at above location = 31152.1 cm^4					
2	66.	5 - 12MM	807.	2000.	NO	YES
	CRITICAL POS MOMENT= 61.73 KN-MET AT 2000.MM, LOAD 318					
	REQD STEEL= 507.MM2, RHO=0.0051, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 42. MMS					
	REQD. DEVELOPMENT LENGTH = 440. MMS					
	Cracked Moment of Inertia Iz at above location = 31152.1 cm^4					
3	334.	5 - 12MM	0.	2000.	YES	YES
	CRITICAL NEG MOMENT= 67.40 KN-MET AT 0.MM, LOAD 314					
	REQD STEEL= 559.MM2, RHO=0.0056, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 42. MMS					
	REQD. DEVELOPMENT LENGTH = 440. MMS					
	Cracked Moment of Inertia Iz at above location = 30746.8 cm^4					

B E A M N O. 104 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 61.68 KNS Vc= 98.10 KNS Vs= 0.00 KNS
Tu= 1.07 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 309
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 167. MM C/C FOR 670. MM

AT END SUPPORT - Vu= 66.23 KNS Vc= 97.98 KNS Vs= 0.00 KNS
Tu= 1.07 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 309
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 167. MM C/C FOR 670. MM



BEAM NO. 105 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 12MM	0.	1500.	YES	YES

CRITICAL POS MOMENT= 11.48 KN-MET AT 0.MM, LOAD 307
REQD STEEL= 122.MM2, RHO=0.0012, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 168. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 14234.5 cm^4

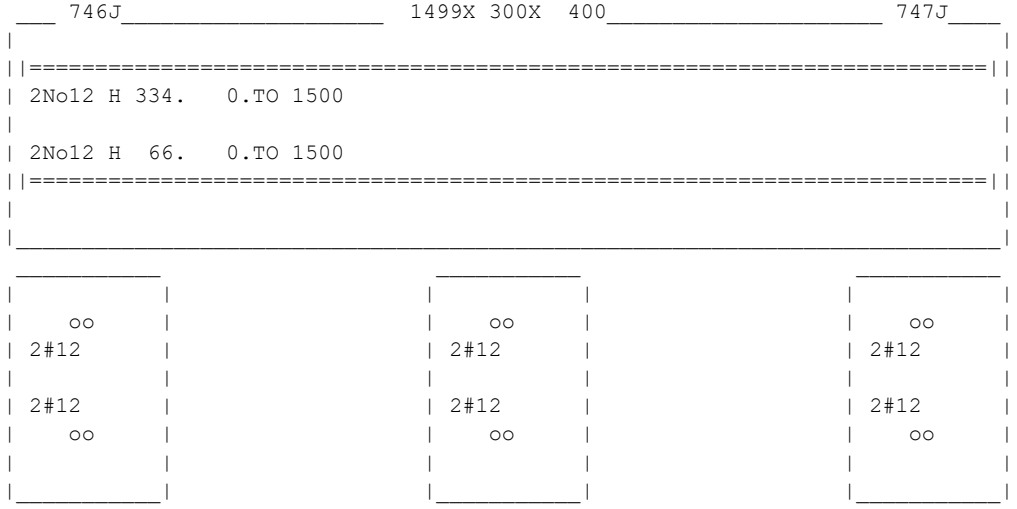
2	334.	2 - 12MM	0.	1500.	YES	YES

CRITICAL NEG MOMENT= 11.81 KN-MET AT 1500.MM, LOAD 311
REQD STEEL= 125.MM2, RHO=0.0013, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 168. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 14234.5 cm^4

B E A M N O. 105 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 0.37 KNS Vc= 94.01 KNS Vs= 0.00 KNS
Tu= 0.22 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 3.70 KNS Vc= 97.78 KNS Vs= 0.00 KNS
Tu= 0.22 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 106 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 12MM	0.	1500.	YES	YES

CRITICAL POS MOMENT= 3.31 KN-MET AT 1500.MM, LOAD 317
REQD STEEL= 35.MM2, RHO=0.0003, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 168. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 14234.5 cm^4

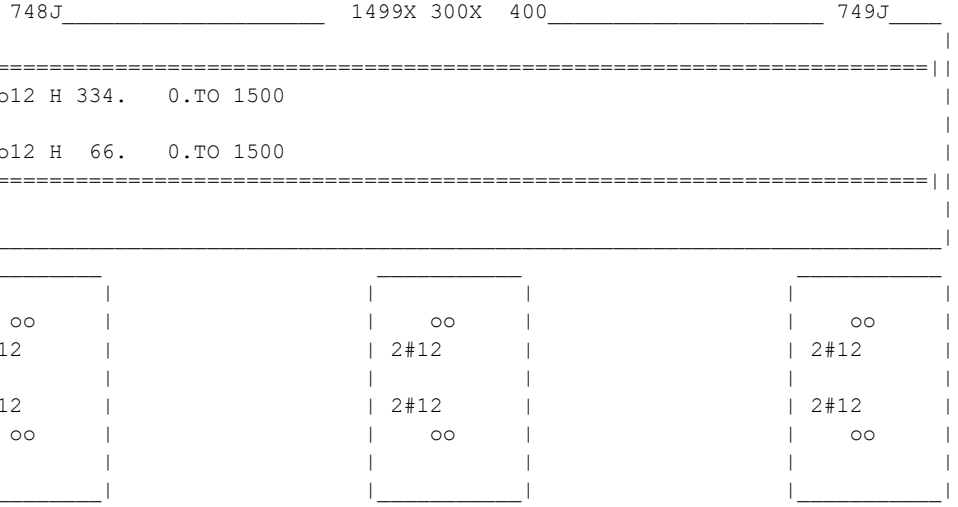
2	334.	2 - 12MM	0.	1500.	YES	YES

CRITICAL NEG MOMENT= 5.60 KN-MET AT 0.MM, LOAD 313
REQD STEEL= 59.MM2, RHO=0.0006, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 168. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 14234.5 cm^4

B E A M N O. 106 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 2.60 KNS Vc= 97.70 KNS Vs= 0.00 KNS
Tu= 0.20 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 0.73 KNS Vc= 172.82 KNS Vs= 0.00 KNS
Tu= 0.20 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

881. END CONCRETE DESIGN
882. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= SEP 2,2020 TIME= 17:32:22 ****

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